

Urine Concentration and Dilution

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CHAPTER OUTLINE

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KEY POINTS

- Nephron segments and vasculature in the renal medulla are arranged in complex but specific anatomic relationships, both in terms of which segment leads to the next segment and in terms of which segments are adjacent to one another, that play an important role in the concentrating and diluting process. The recently discovered interstitial nodal spaces in the inner medulla may participate in this three-dimensional architecture.
- The urinary concentrating mechanism is dependent on two independent processes: (1) generation of a hypertonic medullary interstitium by concentration of NaCl and urea via countercurrent multiplication processes; and (2) osmotic equilibration of the tubule fluid within the medullary collecting ducts with the hypertonic medullary interstitium under the control of vasopressin.
- Vasopressin and the type 2 vasopressin receptor (V₂R) play a central role in the urinary concentrating mechanism. V₂R activation stimulates NaCl reabsorption by the thick ascending limbs of Henle, urea transport in terminal portions of the inner medullary collecting duct (IMCD), and accumulation of the water channel, AQP2, on the apical plasma membrane of collecting duct principal cells.
- Vasopressin binding to V₂R stimulates adenylyl cyclase, predominantly isoform 6, to increase cytosolic cyclic adenosine monophosphate (cAMP) levels as well as intracellular calcium. This stimulates AQP2 accumulation at the apical plasma membrane by inducing depolymerization of the actin cytoskeleton, and by protein phosphorylation, with S256 being an essential site.
- Vasopressin stimulates phosphorylation of the urea transporters (UTs), UT-A1 (at serines 486 and 499), and UT-A3, and their apical plasma membrane accumulation in the inner medullary collecting duct (IMCD) through two cAMP-dependent pathways: protein kinase A (PKA) and Epac (exchange protein activated by cAMP). This leads to increased urea permeability in the IMCD, which facilitates urea reabsorption, increasing medullary interstitial osmolality and the osmotic gradient promoting water reabsorption through AQP2.
- Urea is lost from the inner medullary interstitium, largely via the vasa recta, but urea recycling pathways play a major role in limiting this loss.
- Metformin, an AMP-activated kinase (AMPK) activator, increases UT-A1 and AQP2 phosphorylation and urine-concentrating ability in rodents. Thus, drugs that activate AMPK may be a future therapy for nephrogenic diabetes insipidus.
- Controversy persists as to the nature of the mechanism that generates the inner medullary osmolality, particularly the NaCl gradient, because there is no active NaCl transport in the thin ascending limb. Several recent and ingenious hypotheses have been advanced that depend on the peristalsis of the renal pelvis and the compressibility of the hyaluronan matrix that constitutes the medullary interstitial matrix as integral components of the concentrating mechanism.

INDEPENDENT REGULATION OF WATER AND SALT EXCRETION

The kidney is responsible for numerous homeostatic functions. For example, body fluid tonicity is tightly controlled by the regulation of water excretion, extracellular fluid volume is controlled by regulation of NaCl excretion, systemic acid–base balance is controlled by regulation of net acid excretion, systemic K^+ balance is controlled by regulation of K^+ excretion, and body nitrogen balance¹ is maintained through regulation of urea excretion.

The independent regulation of water and solute excretion is essential for the homeostatic functions of the kidney to be performed simultaneously. This means that in the absence of changes in solute intake or in the metabolic production of waste solutes, the kidney is able to excrete different volumes of water upon changes in water intake. This ability to excrete the appropriate amount of water without marked perturbations in solute excretion (without disturbing the other homeostatic functions of the kidney) is dependent on renal concentrating and diluting mechanisms and forms the basis of this chapter.

Renal water excretion is tightly regulated by the peptide hormone arginine vasopressin (AVP; also named antidiuretic hormone, ADH). Under normal circumstances, the circulating vasopressin level is determined by osmoreceptors in the hypothalamus that trigger increases in vasopressin secretion (by the posterior pituitary gland) when the osmolality of the blood rises above a threshold value, about 292 mOsm/kg H_2O (reviewed by Sands et al.²). This mechanism can be modulated when other inputs to the hypothalamus (e.g., arterial underfilling, severe fatigue, or physical stress) override the osmotic mechanism. Upon an increase in plasma osmolality, vasopressin is secreted from the posterior pituitary gland into the peripheral plasma. The kidney responds to the variable vasopressin levels by varying urine flow (i.e., water excretion). For example, during extreme antidiuresis (high vasopressin), water excretion is greater than 100-fold lower than during major water diuresis (low vasopressin). These major changes in water excretion are obtained without substantial changes in steady-state solute excretion (Fig. 10.1). This phenomenon is dependent on the kidney's ability to concentrate and dilute the urine. During low circulating vasopressin levels, urine osmolality is less than that of plasma (290 mOsm/kg H_2O): the diluting function of the kidney. In contrast, when the circulating vasopressin level is high, urine osmolality is much higher than that of plasma: the concentrating function of the kidney.

ORGANIZATION OF STRUCTURES IN THE KIDNEY RELEVANT TO URINARY CONCENTRATING AND DILUTING PROCESS

The kidney's ability to vary water excretion over a wide physiologic range, without altering steady-state solute excretion, cannot be simply explained as a consequence of the sequential transport processes along the nephron.³ The independent regulation of water and sodium excretion occurs in the renal medulla, where the nephron segments and

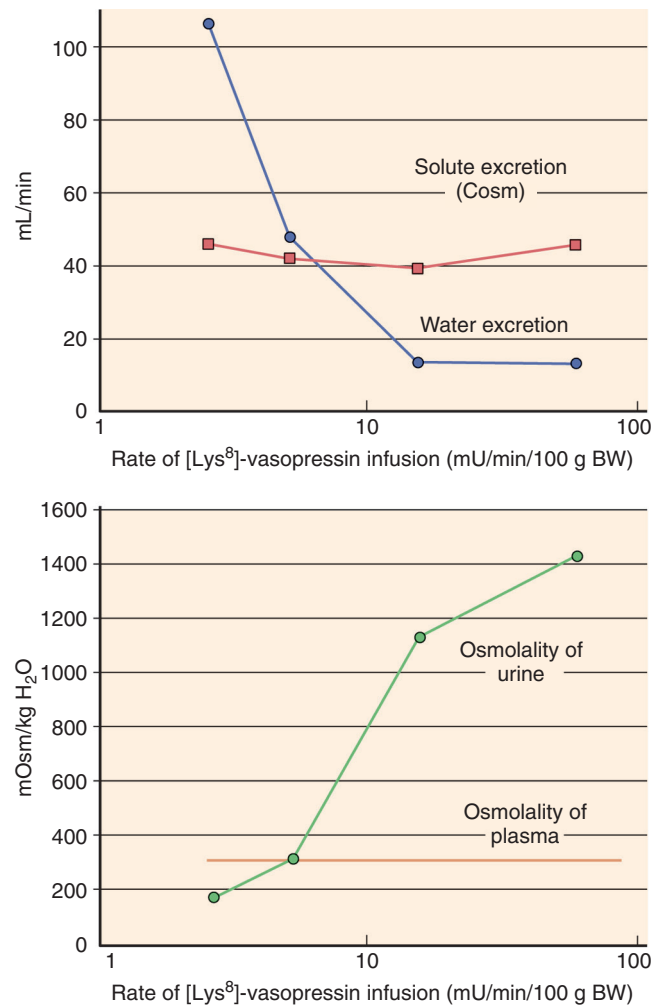


Fig. 10.1 Steady-state renal response to varying rates of vasopressin infusion in conscious rats. A water load (4% of body weight) was maintained throughout the experiments to suppress endogenous vasopressin secretion. Although the urine flow rate was markedly reduced at higher vasopressin infusion rates, the osmolar clearance (solute excretion) changed little. Concordantly, at higher vasopressin infusion rates, the osmolality of the urine increases significantly, whereas plasma osmolality remains constant. BW, Body weight. Data from Atherton JC, Hai MA, Thomas S. The time course of changes in renal tissue composition during water diuresis in the rat. *J Physiol.* 1068; 197:429–443.

vasculature (vasa recta) are arranged in complex but specific anatomic relationships, both in terms of which segments connect to which segments and their three-dimensional configuration. Thus, it is necessary to consider the parallel interactions between nephron segments that occur as a result of its looped or hairpin structure. Fig. 10.2 illustrates the regional architecture of the renal medulla and medullary rays.⁴

Fig. 10.3 shows a schematic representation of the mammalian nephron with the localization of major water channels (aquaporins; AQP), urea transporters (UTs), and ion transporters important to the urinary concentrating process. Fig. 10.4 shows which of these transporters and channels are molecular targets for regulated vasopressin action, either in abundance or activity, and thus likely to play a role in urine

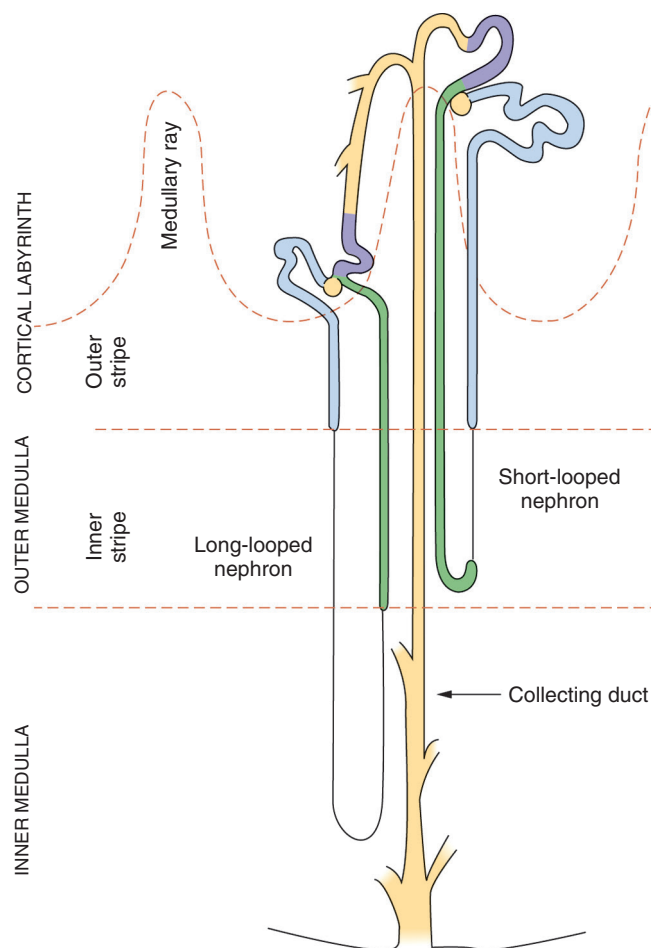


Fig. 10.2 Mammalian renal structure. Major regions of the kidney are shown on the left. Configurations of a long-looped and a short-looped nephron are depicted. The major portions of the nephron are proximal tubules (medium blue), thin limbs of loops of Henle (single line), thick ascending limbs of loops of Henle (green), distal convoluted tubules (lavender), and the collecting duct system (yellow). Modified from Knepper MA, Stephenson JL. Urinary concentrating and diluting processes. In: Andreoli TE, Fanestil DD, Hoffman JF, Schultz SG, eds. *Physiology of Membrane Disorders*. New York: Plenum; 1986:713–726.

concentration. The functions of several of the transporters and channels shown in Fig. 10.3 have been evaluated in mice using gene deletion techniques (reviewed by Fenton et al.⁵). The phenotypes of these mice have been informative with regard to the role of these proteins and their nephron segments in the urinary concentrating and diluting mechanisms.

RENAL TUBULE

LOOPS OF HENLE

The kidney generally contains two populations of nephrons, long-looped and short-looped, which merge to form a common collecting duct system (see Fig. 10.2). Both types of nephrons have loops of Henle that are arranged in a folded or hairpin configuration. Short-looped nephrons generally have glomeruli that are located more superficially in the cortex and have loops that bend in the outer medulla. Long-looped nephrons generally have glomeruli that are located more deeply within the cortex and have loops that

bend at various levels of the inner medulla. Long-looped nephrons also contain a thin ascending limb, a segment that is not present in short-looped nephrons. Thin ascending limbs are found only in the inner medulla. The inner-outer medullary border is defined by the transition from thin to thick ascending limbs. Thus, the outer medulla contains only thick ascending limbs, regardless of the type of loop. The long-looped nephrons bend at various levels of the inner medulla from the inner-outer medullary border to the papillary tip. Thus, progressively fewer loops of Henle extend to deeper levels of the inner medulla. Some mammalian kidneys, such as human kidneys, also contain cortical nephrons, which are nephrons whose loops of Henle do not reach into the medulla.

The loops of Henle receive tubular fluid from the proximal convoluted tubules. Tubular fluid exits the thick ascending limbs of both long- and short-looped nephrons, and from cortical nephrons in species that have them, and flows into distal convoluted tubules. Thus, the descending and ascending limbs of the loops of Henle have a countercurrent flow configuration and are composed of several different nephron segments (see Fig. 10.2). The descending portion of the loop of Henle consists of the S2 proximal straight tubule in the medullary ray, the S3 proximal straight tubule (or pars recta) in the outer stripe of the outer medulla, and the thin descending limb in the inner stripe of the outer medulla and the inner medulla. The descending thin limb of short-looped nephrons differs structurally and functionally from the descending thin limb of long-looped nephrons.^{6,7}

The location of the descending thin limb of short-looped nephrons within the outer medulla is illustrated in Fig. 10.5 (labeled in green).⁸ The descending thin limbs of short-looped nephrons surround the vascular bundles in the outer medulla and tend to be organized in a ring-like pattern (see Fig. 10.5, inset). Thin descending limbs of long-looped nephrons in the outer medulla differ morphologically and functionally from thin descending limbs of long-looped nephrons in the inner medulla.^{9–12} The histologic transition from the outer medullary to the inner medullary type of thin descending limbs of long-looped nephrons is gradual and often occurs at some distance into the inner medulla, rather than strictly at the inner-outer medullary border as is the case for the transition between thin and thick ascending limbs.

Pannabecker and coworkers used immunohistochemical labeling and computer-assisted reconstruction to provide new detail about the functional architecture of the rat inner medulla.^{13–15} Fig. 10.6 shows a computerized reconstruction of the inner medullary portion of several long-looped nephrons from rats that are labeled using antibodies to the water channel aquaporin-1 (AQP1, shown in red) and the chloride channel ClC-K1 (shown in green) (reviewed by Pannabecker et al.^{13–16}). AQP1 is a marker of thin descending limbs of long-looped nephrons in the outer medulla, and it is detected in thin descending limbs of long-looped nephrons in the inner medulla for a variable distance. However, AQP1 was not found in the thin descending limbs of the loops of Henle that turn within the upper millimeter of the inner medulla. Correspondingly, Zhai and colleagues determined that AQP1 was not detectable along the entire length of thin descending limbs of short-looped rat nephrons.¹⁷ In contrast, the upper 40% of thin descending limbs that turn below the first millimeter express AQP1, whereas the lower 60% do

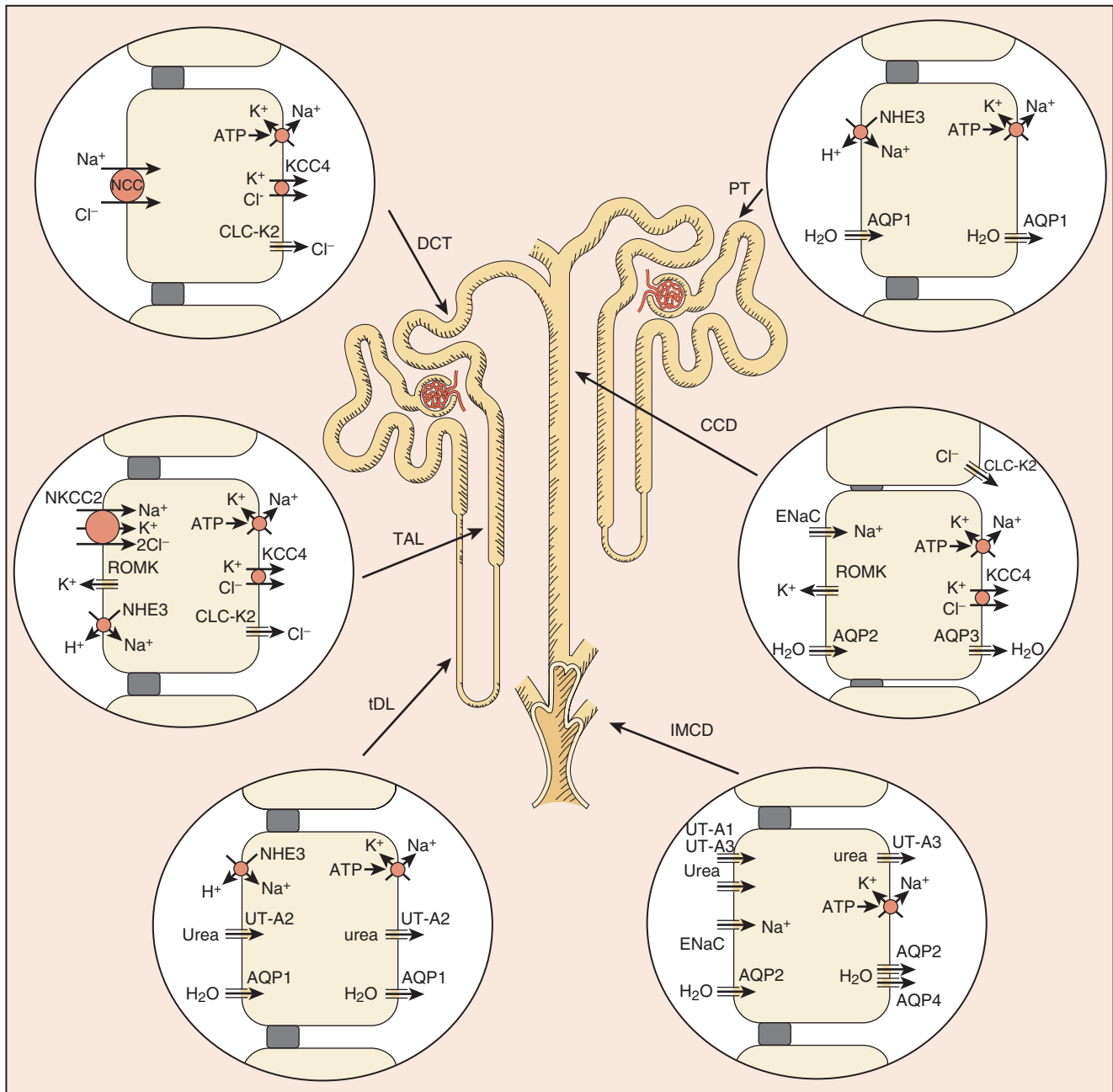


Fig. 10.3 Major aquaporins, urea transporters, and ion transporters/channels that are important to the urinary concentrating and diluting process. Figure depicts a schematic overview of a mammalian kidney tubule, showing the solute and water transport pathways in the proximal tubule (PT), thin descending limb of Henle loop (tDL), thick ascending limb (TAL), distal convoluted tubule (DCT), cortical collecting duct (CCD), and inner medullary collecting duct (IMCD). Tubule lumen side is always on the left-hand side of the cell, whereas the interstitium is on the right-hand side. Arrows represent direction of movement. Adapted from Fenton RA, Knepper MA. Mouse models and the urinary concentrating mechanism in the new millennium. *Physiol Rev.* 2007;87:1083–1112.

not. ClC-K1 is a marker of the thin ascending limb-type epithelium. It is first detected just before the bend of the loops of Henle, consistent with several morphologic studies demonstrating that the descending limb to ascending limb demarcation occurs before the loop bend. A substantial portion of the inner medullary thin descending limb of long-looped nephrons did not express either AQP1 or ClC-K1, as indicated in gray in Fig. 10.6.

The deepest portions of descending thin limbs have low water permeability and reduced AQP1.¹⁸ It has been proposed,

but not demonstrated experimentally, that urine concentration would be improved by the presence of a urea- Na^+ or urea- Cl^- cotransporter in the AQP1-null portion of the thin descending limb.¹⁹ These deep AQP1-null, prebend segments and thin ascending limbs lie equally near the collecting ducts.¹⁸ However, the distal 30% of thin ascending limbs of the longest loops of Henle lie distant from collecting ducts.¹⁸ Urea permeability is lower in the upper portion of thin descending limbs than in the lower portion or in thin ascending limbs in Munich-Wistar rats.²⁰ Because phloretin does not inhibit urea

	LDL-OM	LDL-IM	SDL (early)	SDL (late)	ATL	MTAL	CTAL	DCT	CNT	ICT	CCD	OMCD	IMCD (initial)	IMCD (terminal)
Aquaporin-1	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin											
Aquaporin-2									Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin
Aquaporin-3									Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin		
Aquaporin-4												Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin
Urea transporter UT-A1														Expressed, regulated by vasopressin
Urea transporter UT-A2		Expressed, regulated by vasopressin		Expressed, regulated by vasopressin										
Urea transporter UT-A3														Expressed, regulated by vasopressin
Na-H exchanger (NHE3)	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin				Expressed, regulated by vasopressin	Expressed, regulated by vasopressin							
Na-K-2Cl cotransporter (NKCC2)						Expressed, regulated by vasopressin	Expressed, regulated by vasopressin							
Na-K-2Cl cotransporter (NKCC1)													Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin
Na-Cl cotransporter (NCC)								Expressed, regulated by vasopressin						
Epithelial Na channel (ENaC)									Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin
Cl channel CIC-K1					Expressed, regulated by vasopressin									
Cl channel CIC-K2						Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin
Potassium channel (ROMK)						Expressed, regulated by vasopressin	Expressed, regulated by vasopressin		Expressed, regulated by vasopressin	Expressed, regulated by vasopressin	Expressed, regulated by vasopressin			
K-Cl cotransporter (KCC4)						Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin	Expressed, not regulated by vasopressin					

Fig. 10.4 Grid showing sites of expression of water channels, urea transporters, and ion transporters important to the urinary concentrating process. (See text for details).

permeability, urea transport is not mediated by the UT-A2 urea transporter in these segments.²⁰ Two novel variants of UT-A2, UT-A2c, and UT-A2d, and a variant of the sodium-glucose cotransporter 1, SGLT1a, are expressed in the lower portion of thin descending and thin ascending limb segments, and may mediate urea transport.²¹

Pannabecker and Dantzer²² identified three population groups of loops of Henle in Munich-Wistar rats that can be

distinguished by the position of the thin ascending limb at the base of the inner medulla and by differing loop length (Fig. 10.7). Group 1 loops have thin ascending limbs that are interposed between collecting ducts; group 2 loops have thin ascending limbs that are adjacent to just one collecting duct; and group 3 loops have thin ascending limbs that lie more than 0.5 tubule diameters from a collecting duct. As the collecting ducts coalesce and the shorter loops of Henle

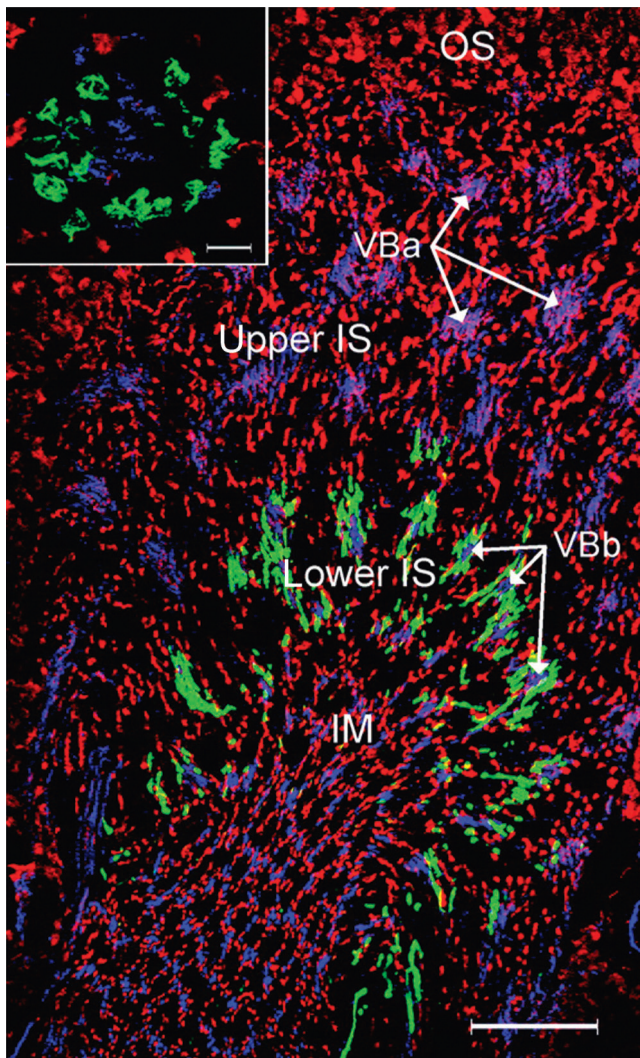


Fig. 10.5 Triple immunolabeling of rat renal medulla showing localization of UT-A2 (green), marking late thin descending limbs from short-looped nephrons, von Willebrand factor (blue) marking endothelial cells of vasa recta, and aquaporin-1 (red) marking thin descending limbs from outer medullary long-looped nephrons and early short-looped nephrons. Inset shows a cross-section of a vascular bundle demonstrating that UT-A2-positive thin descending limbs from short-looped nephrons surround the vascular bundles in the deep part of the outer medulla. IM, Inner medulla; IS, inner stripe of outer medulla; OS, outer stripe of outer medulla; VBa, vascular bundles in outer part of inner stripe; VBb, vascular bundles in inner part of inner stripe. Reproduced with permission from Wade JB, Lee AJ, Liu J, et al. UT-A2. A 55 kDa urea transporter protein in thin descending limb of Henle loop whose abundance is regulated by vasopressin. *Am J Physiol Renal Physiol.* 2002;278:F52–F62.

disappear, the originating portions of the longer thin ascending limbs run alongside the collecting ducts for a substantial distance.²²

Detailed studies of inner medullary structure, both by Kriz and colleagues^{23–26} and more recently by Pannabecker and colleagues,^{16,19,22,27–30} found that the inner medullary collecting ducts in the inner medullary base (initial inner medullary collecting ducts) form clusters that coalesce along the cortico-medullary axis. The thin descending limbs are

predominantly present at the periphery of these clusters and appear to form an asymmetric ring around each collecting duct cluster, whereas the thin ascending limbs are distributed relatively uniformly among the collecting ducts and thin descending limbs.^{27,31}

In the rat, each collecting duct is surrounded by approximately four ascending vasa recta.³² One or two thin ascending limbs lie between each ascending vasa recta, and opposite to the collecting duct.³² Pannabecker and colleagues hypothesized that descending and ascending thin limbs enter and exit collecting duct clusters in a manner that is important for the generation and maintenance of the osmolality gradient within the inner medulla.¹⁸ These structures form an interstitial nodal space that runs axially through the inner medulla and that may carry water, urea, and NaCl.³² These anatomic relationships may facilitate the preferential mixing of solutes and fluid within the interstitial nodal space.³³ In humans, interstitial nodal spaces are relatively infrequent.³⁰

Kidney-Specific Chloride Channel 1 (ClC-K1)

ClC-K1 localizes to both the apical and basolateral plasma membranes of thin ascending limbs.³⁴ Additionally, ClC-K1 mRNA has been detected in both the thick ascending limb and distal convoluted tubule.³⁵ In isolated perfused tubules, the chloride conductance of thin ascending limbs is increased by vasopressin exposure: either as a result of increased unit conductance or altered cellular localization of ClC-K1 chloride channels.³⁶ Microperfusion studies of ClC-K1 null mice (*Clcnk1*^{−/−}) determined that there was drastically reduced transepithelial chloride transport in the thin ascending limbs of knockout mice.³⁷ *Clcnk1*^{−/−} mice had significantly greater urine volume and lower urine osmolality compared with controls, and even after water deprivation or vasopressin administration, knockout mice were unable to concentrate their urine. This observed polyuria was due to water diuresis and not osmotic diuresis. Inner medullary concentrations of Na⁺ and Cl[−] from *Clcnk1*^{−/−} mice were approximately half those of controls, resulting in a significantly reduced osmolality of the papilla. These studies demonstrate that ClC-K1 is necessary for maintenance of maximal osmolality in the inner medullary tissue. The findings in the *Clcnk1*^{−/−} mice emphasize the importance of rapid chloride exit (and presumably sodium exit) from thin ascending limbs in the inner medullary concentrating process and provide support for the “passive mechanism” (see later).

Na-K-2Cl Cotransporter Type 2 (NKCC2) and Na⁺-H⁺-Exchanger Isoform Type 3 (NHE3)

NKCC2 and NHE3 are the major apical transporters mediating Na⁺ entry in the thick ascending limb.^{38–41} However, knockout of NKCC2 or NHE3 results in drastically different effects on the urinary concentrating mechanism.^{42,43} Total NHE3 knockout mice have a marked reduction in proximal tubule fluid absorption, with a compensatory decrease in glomerular filtration rate owing to an intact tubuloglomerular feedback mechanism.⁴⁴ On *ad libitum* water intake, total NHE3 knockout mice manifest a moderate increase in water intake associated with lower urinary osmolality.⁴⁵ In addition, renal tubule-selective NHE3 knockout mice have only small increases in fluid intake and urinary flow under basal conditions and a minor urinary concentrating defect.⁴⁶ In contrast, NKCC2 knockout mice die before weaning due to renal fluid wasting

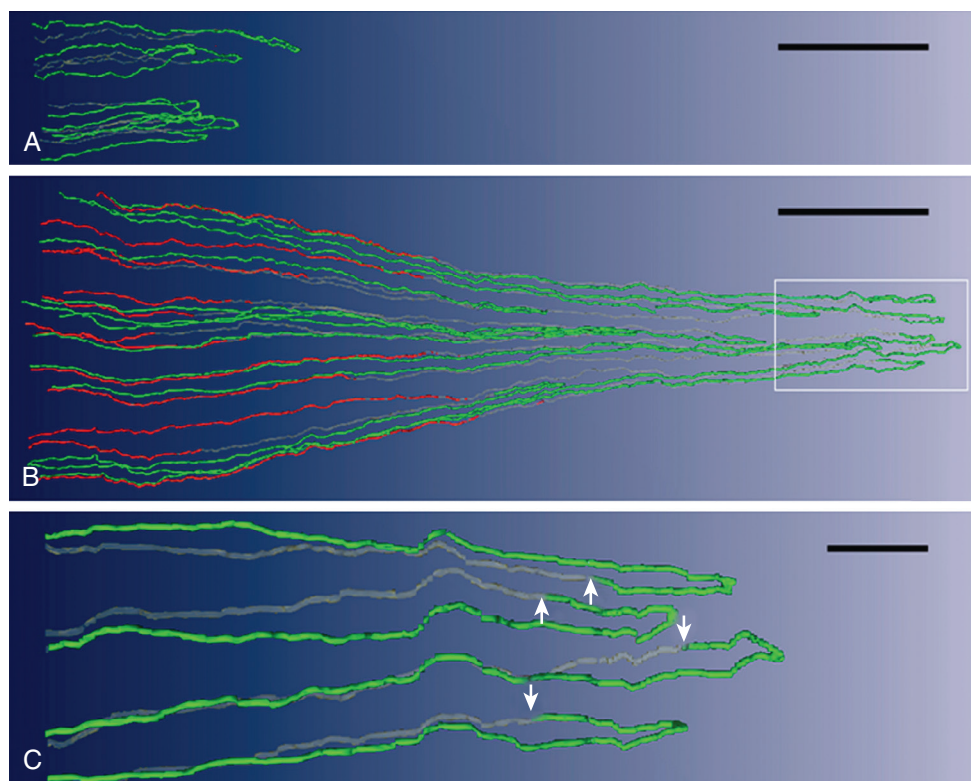


Fig. 10.6 Computer-assisted reconstruction of loops of Henle from rat inner medulla showing expression of aquaporin-1 (AQP1; red) and CIC-K1 (green); gray regions (B-crystallin) express undetectable levels of AQP1 and CIC-K1. Loops are oriented along the corticopapillary axis, with the left edge of each image nearer the base of the inner medulla. (A) Thin limbs that have their bends within the first millimeter beyond the outer-inner medullary boundary. Descending segments lack detectable AQP1. CIC-K1 is expressed continuously along the prebend segment and the thin ascending limb. (B) Loops that have their bends beyond the first millimeter of the inner medulla. AQP1 is expressed along the initial 40% of each thin descending limb and is absent from the remainder of each loop. CIC-K1 is expressed continuously along the prebend segment and the thin ascending limb. Boxed area is enlarged in (C). (C) Enlargement of near-bend regions of four thin limbs from box in (B). CIC-K1 expression, corresponding to thin descending limb prebend segment, begins, on average, 165 μm before the loop bend (arrows). Scale bars, 500 μm (A) and (B) and 100 μm (C). Reproduced with permission from: Pannabecker TL, Dantzer WH, Layton HE, et al. Role of three-dimensional architecture in the urine concentrating mechanism of the rat renal inner medulla. *Am J Physiol Renal Physiol.* 2008;295:F1271–F1285.

and dehydration,⁴³ highlighting the essential role of NKCC2 in the urinary concentrating mechanism.

Why does the deletion of NKCC2 result in such a severe phenotype, when the deletion of NHE3, a transporter responsible for reabsorption of far more Na^+ , results in a viable mouse capable of maintaining extracellular fluid volume? The answer appears to be in the special role that NKCC2 plays in the macula densa in the mediation of tubuloglomerular feedback. Tubuloglomerular feedback allows NHE3 knockout mice to maintain a relatively normal distal delivery through a decrease in glomerular filtration rate, whereas NKCC2 mice cannot compensate in this manner because the transporter is necessary for tubuloglomerular feedback to occur.^{47,48}

Renal Outer Medullary Potassium Channel (ROMK, Kir 1.1)

ROMK is an ATP-sensitive inwardly rectifier potassium channel that localizes to the thick ascending limb, distal convoluted tubule, connecting tubule, and collecting duct system, where it is predominantly associated with the apical plasma membrane.^{49–53} Chronic vasopressin treatment increases

ROMK abundance in thick ascending limbs, thus contributing to vasopressin's long-term effect to increase NaCl transport in this segment.^{54,55} The majority of ROMK knockout mice die before weaning due to hydronephrosis and severe dehydration.⁵⁶ Although 5% of these mice survive the perinatal period, adult mice manifest polydipsia, polyuria, impaired urinary concentrating ability, hypernatremia, and reduced blood pressure, consistent with the known role of ROMK in active NaCl absorption in the thick ascending limb. From these animals, a line of mice has been derived that has a greater survival rate and no hydronephrosis in adult animals; yet the urine concentrating defect persists.

DISTAL TUBULE SEGMENTS IN THE CORTICAL LABYRINTH

After tubular fluid exits the loop of Henle through the cortical thick ascending limb, it enters the distal convoluted tubule, which is located in the cortical labyrinth. In most mammalian species, several distal tubules merge to form a connecting tubule arcade.⁵⁷ The connecting tubule cells express both the vasopressin-regulated water channel, aquaporin-2 (AQP2), and the type 2 vasopressin receptor (V_2R),⁵⁸ suggesting that

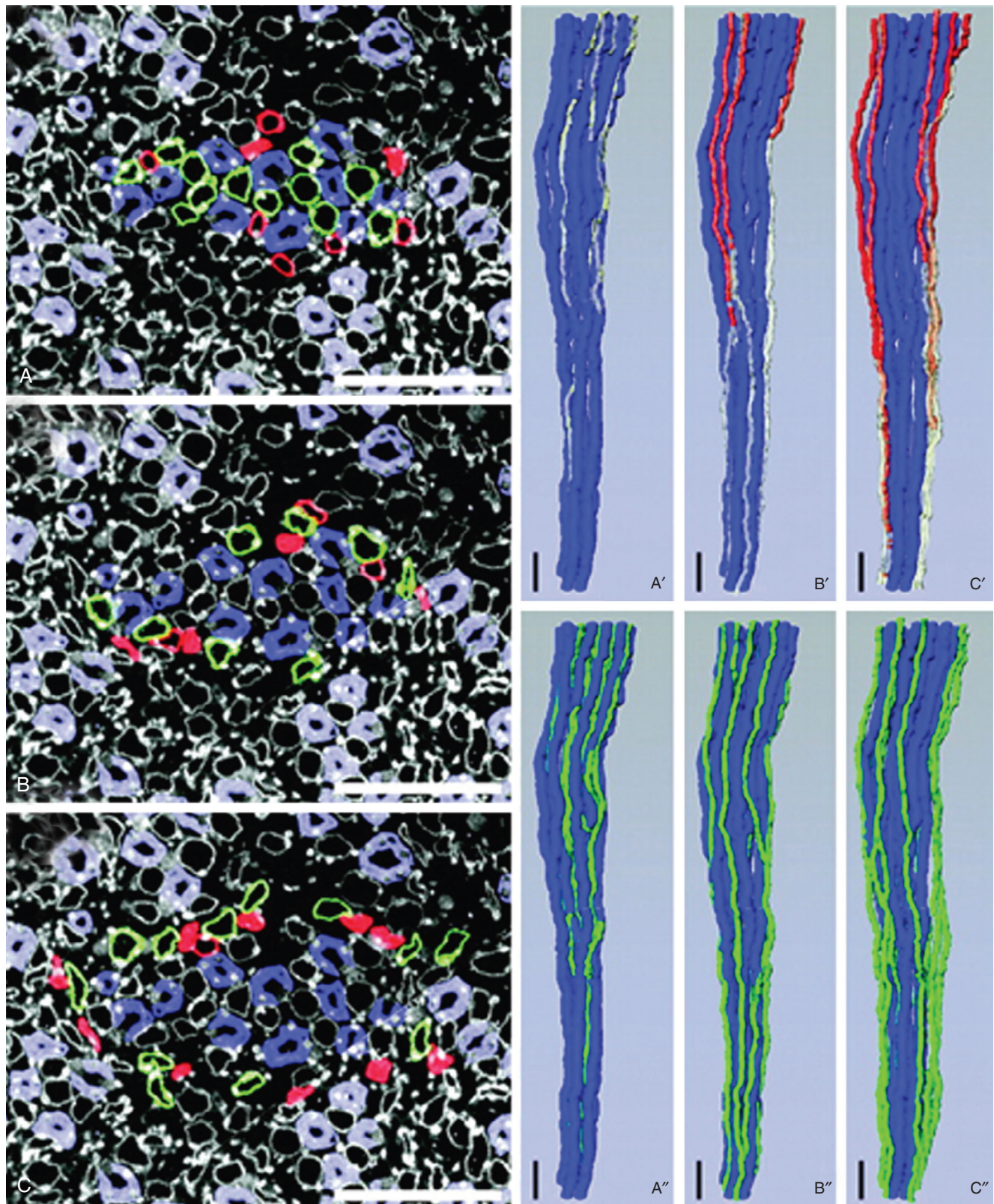


Fig. 10.7 Spatial relationships between thin descending limbs (*red tubules*), thin ascending limbs (*green tubules*), and collecting ducts (*dark blue tubules*). Thin ascending limbs were categorized into three groups related to their lateral proximity to collecting ducts. Members of each group are shown in a transverse section located at the base of the inner medulla: (A) group 1; (B) group 2; and (C) group 3. In (A), (B), and (C), *open red figures* represent aquaporin-1 (AQP1)-null thin descending limbs, *solid red figures* represent AQP1-expressing thin descending limbs, *white outlined figures* represent thin ascending limbs not associated with the collecting duct cluster, and *light blue figures* represent collecting ducts not associated with the collecting duct cluster. Two prebend segments from group 1 are included in (A). One thin ascending limb from each of groups 2 and 3 (B) and (C) extends below the region of reconstruction, and their thin descending limbs were therefore not reconstructed. (A'), (B'), and (C') show thin descending limbs and collecting ducts; (A''), (B''), and (C'') show thin ascending limbs and collecting ducts. Gray tubules in (A'), (B'), and (C') represent AQP1-null thin descending limbs. Scale bars, 100 μ m. Reproduced with permission from Pannabecker TL, Dantzler WH. Three-dimensional lateral and vertical relationships of inner medullary loops of Henle and collecting ducts. *Am J Physiol Renal Physiol*. 2004;287:F767–F774.

the arcades are sites of vasopressin-regulated water reabsorption, similar to collecting ducts (see later). Tubular fluid exits the connecting tubules within the arcades and enters the initial collecting tubules, located in the superficial cortex, and then into the cortical collecting ducts. In most rodent species that have been studied, several nephrons merge to form a single cortical collecting duct.^{6,59}

COLLECTING DUCT SYSTEM

The collecting duct system spans all regions of the kidney, starting in the cortex and running to the tip of the inner medulla (see Fig. 10.2). The collecting ducts are the major site of vasopressin-regulated water and urea transport. The transport of water and urea is crucial to the urine-concentrating mechanism, and these are discussed in detail later in this chapter. The collecting ducts are arranged in parallel to the loops of Henle in the medullary rays, outer medulla, and inner medulla. Like the loops of Henle, several morphologically and functionally discrete segments are contained within the collecting duct system. In general, the collecting ducts descend straight through the medullary rays and outer medulla without joining with other collecting ducts. However, several collecting ducts merge as they descend within the inner medulla, resulting in a progressive reduction in the number of inner medullary collecting ducts from the inner-outer medullary border to the papillary tip.⁵⁹ The tapered structure of the renal papilla results from the reduction in collecting duct number, accompanied by a progressive reduction in the number of loops of Henle, reaching the deepest levels of the inner medulla.

The epithelial sodium channel (ENaC) is localized to the late distal convoluted tubule, connecting tubule, initial collecting tubule, and throughout the collecting duct.^{60,61} Vasopressin treatment increases the protein abundance of the beta- and gamma-subunits of ENaC.^{62–64} Acute vasopressin exposure also increases Na⁺ reabsorption in the cortical collecting duct by increasing apical Na⁺ entry via ENaC,^{65–67} due to adenylyl cyclase (AC) 6-dependent stimulation of ENaC open probability and apical membrane channel number.⁶⁸ Deletion of any of the ENaC subunits results in a severe phenotype with neonatal death.^{69–72} Alpha ENaC deletion from the collecting ducts alone, leaving intact ENaC expression in the connecting tubule and nonrenal tissues, results in viable mice that have little or no difficulty in maintaining salt and fluid homeostasis.⁷³ In contrast, alpha ENaC deletion from the connecting tubule and collecting duct together results in a mouse model with increased urine volume and decreased urine osmolality,⁷⁴ indicating that alpha ENaC expression within the connecting tubule and collecting duct is crucial for sodium and water homeostasis.

VASCULATURE

For detailed description of the renal vasculature, see Chapter 2 (Anatomy of the Kidney). The major blood vessels that carry blood into and out of the renal medulla are named the vasa recta. Blood enters the descending vasa recta from the efferent arterioles of juxtamedullary nephrons and supplies it to the capillary plexuses at each level of the medulla. The outer medullary capillary plexus is denser and better perfused than the plexus in the inner medulla.⁷⁵ Blood from the inner medullary capillary plexus feeds into the ascending

vasa recta (ascending vasa recta are never formed directly from descending vasa recta in a loop-like structure). Inner medullary ascending vasa recta traverse the inner stripe of the outer medulla in close physical association with the descending vasa recta in vascular bundles.²³ In many animal species, thin descending limbs of short-looped nephrons surround the vascular bundles, as shown in Fig. 10.5. Here the thin descending limb segments are labeled with an antibody to the UT-A2 urea transporter,⁸ suggesting a route for urea recycling from the vasa recta to the thin descending limbs of short-looped nephrons. The outer medullary capillary plexus is drained by vasa recta that ascend through the outer stripe of the outer medulla, separate from the descending vasa recta.²⁶ Recent computer-assisted digital tracing of the mouse kidney combined with AQP1 immunohistochemistry shows that the arrangement of tubules and vessels in the vascular bundles is important for providing a pathway for lateral osmolality heterogeneity for urine concentration.⁷⁶

The counterflow arrangement of the vasa recta in the medulla promotes countercurrent exchange of solutes and water, which is facilitated by the presence of AQP1^{77,78} and UT-B UTs^{79–81} in the endothelial cells of the descending portion of the vasa recta. In rats, UT-B is expressed in both the outer medullary and inner medullary descending vasa recta.^{81,82} In humans, UT-B is also expressed in the descending vasa recta, but its expression decreases with depth in the inner medulla.³⁰

Countercurrent exchange provides a means of reducing the effective blood flow to the medulla while maintaining a high absolute perfusion rate.⁸³ The low effective blood flow that results from countercurrent exchange is thought to be important for the preservation of solute concentration gradients in the medullary tissue (see later).

In contrast to the medulla, the cortical labyrinth has a high effective blood flow. The rapid vascular perfusion to this region promotes the rapid return of solutes and water reabsorbed from the nephron to the general circulation. The rapid perfusion is thought to maintain the interstitial concentrations of most solutes at levels close to those in the peripheral plasma. The medullary rays of the cortex have a capillary plexus that is considerably sparser than that of the cortical labyrinth. Consequently, the effective blood flow to the medullary rays has been postulated to be lower than that of the cortical labyrinth.³

MEDULLARY INTERSTITIUM

The renal medullary interstitium connects the tubules and vasculature.⁸⁴ It is a complex space that includes the medullary interstitial cells, microfibrils, extracellular matrix, and fluid.^{31,84–86} The interstitium is relatively small in volume in the outer medulla and the outer portion of the inner medulla, which may be important in limiting the diffusion of solutes upward along the medullary axis.^{3,27,84} In contrast, the interstitial space is much larger in the inner half of the inner medulla.^{3,27,84} Within this region, it consists of a gelatinous matrix containing large amounts of highly polymerized hyaluronic acid, consisting of alternating N-acetyl-D-glucosamine and D-glucuronate moieties.⁸⁷ Theories have been proposed in which the hyaluronic acid interstitial matrix plays a direct role in the generation of an inner medullary osmotic gradient through its ability to store and transduce energy

from the smooth muscle contractions of the renal pelvis (see later).⁸⁷

RENAL PELVIS

Urine exits the collecting duct system through the ducts of Bellini at the papillary tip and enters the renal pelvis (Fig. 10.8). The renal pelvis (or calyx in multipapillate kidneys) is a complex intrarenal urinary space that surrounds the papilla. The renal pelvis has portions that extend into the outer medulla, which are called fornices and secondary pouches. Although a transitional epithelium lines most of the pelvic space, the renal parenchyma is separated from the pelvic space by a simple cuboidal epithelium.⁸⁸ In humans, the UT-B urea transporter is expressed within this papillary surface epithelium.⁵⁰ It has been proposed that water and solute transport could occur across this epithelium, thereby modifying the composition of the renal medullary interstitial fluid.⁸⁹ There are two smooth muscle layers within the renal pelvic (calyceal) wall.⁹⁰ Contractions of these smooth muscle layers generate powerful peristaltic waves that appear to displace the renal papilla downward with a “milking” action.⁹¹ These peristaltic waves may intermittently propel urine along the collecting ducts. The contractions compress all structures within the renal inner medulla, including the interstitium, loops of Henle, vasa recta, and collecting ducts.⁹² Theories have been proposed whereby these contractions furnish part of the energy for concentrating solutes, and hence concentrating urine, within the inner medulla (see later).⁸⁷

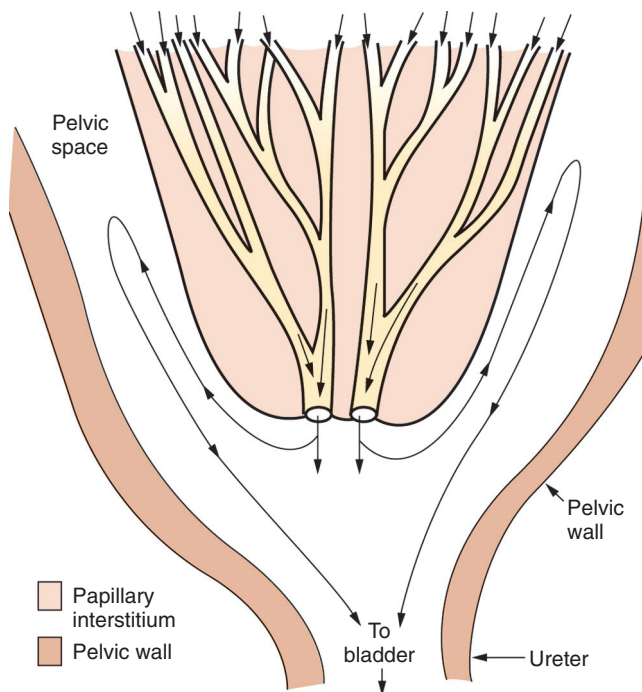


Fig. 10.8 Pattern of urine flow in papillary collecting ducts and renal pelvis. Urine exits the papillary collecting ducts (ducts of Bellini) at the tip of the renal papilla and is carried to the urinary bladder by the ureter. Under some circumstances, a fraction of the urine may reflux backward in the pelvic space and contact the outer surface of the renal papilla. Solute and water exchange across the papillary surface epithelium has been postulated (see text).

VASOPRESSIN AND THE TYPE 2 VASOPRESSIN RECEPTOR

The small peptide hormone vasopressin and the V_2R play a central role in the urinary concentrating mechanism. V_2R activation stimulates NaCl reabsorption by the thick ascending limbs of Henle, urea transport in terminal portions of the inner medullary collecting duct, and accumulation of a water channel, AQP2, on the plasma membrane of collecting duct principal cells. These events permit the collecting duct luminal fluid to equilibrate osmotically with the surrounding interstitium in the kidney, resulting in water reabsorption and urine concentration. Dysfunction of this reabsorption mechanism in the collecting duct results in the production of large amounts of dilute urine, up to 18 L/d, a disease known as diabetes insipidus. In the following we address how the V_2R and AQP2 interact via intracellular signaling pathways to regulate collecting duct water reabsorption and urine concentration.

Clinical Relevance

Nephrogenic Diabetes Insipidus

Nephrogenic diabetes insipidus (NDI) results from the inability of the kidney to respond to vasopressin and produce a concentrated urine. Congenital NDI results from mutations in the V_2R in 90% of families (in which the mutation is known) and in AQP2 in most of the other 10% (reviewed in Sands and Bichet⁴⁶⁹). Acquired forms of NDI occur much more frequently and arise as a consequence of drug treatments, electrolyte disturbances, and urinary tract obstruction. In most manifestations of acquired NDI, dysregulation of AQP2, either in terms of protein abundance or in AQP2 membrane targeting, plays a fundamental role in the development of polyuria.^{470,471} Downregulation of AQP2 observed in acquired NDI is most likely the primary cause of the NDI, rather than being a secondary event (e.g., as a consequence of the increased urine production or reduction in interstitial osmolality). For example, in models of hypokalemic and lithium-induced NDI, the changes in AQP2 expression in the kidney cortex are identical to those seen in the inner medulla,^{472–474} which indicates that interstitial tonicity is not a major factor. Moreover, washout of the medullary osmotic gradient for 1 or 5 days using the loop diuretic furosemide has no effect on AQP2 expression,^{474,475} which indicates that high urine flow in itself is not responsible for the reduced AQP2 expression in experimental NDI. Studies investigating the molecular physiology and signaling pathways regulating water and urea transport have identified several novel therapeutic possibilities for treating NDI (reviewed in Sands and Klein⁴⁰³).

VASOPRESSIN

The ADH of most mammals is a nine-amino acid peptide, vasopressin. Secretion of vasopressin from the posterior pituitary is stimulated by an increase in plasma osmolality,

but also by a reduction in plasma volume (reviewed in Sands et al.²). Vasopressin activates regulatory systems necessary to retain water and restore osmolality to normal.⁹³ The effects of vasopressin occur through the stimulation of receptors that are located on different cell types.^{94,95} Here we focus on the V_2R activation of a cyclic adenosine monophosphate (cAMP) pathway in renal epithelial cells for modulation of collecting duct water transport.

TYPE 2 VASOPRESSIN RECEPTOR

The V_2R is a seven transmembrane-spanning domain receptor that couples to heterotrimeric G proteins (GPCRs).^{96,97} In the kidney it is expressed from the thick ascending limb of

the loop of Henle through to the collecting duct principal cells.⁹⁸⁻¹⁰² When vasopressin binds to the V_2R , AC activity is stimulated and cytosolic cAMP levels increase.¹⁰³ This ultimately leads to an accumulation of AQP2 in the apical plasma membrane of collecting duct principal cells, thus increasing transepithelial water permeability and facilitating osmotically driven water reabsorption (Fig. 10.9). Intracellular calcium is also increased by vasopressin via a mechanism involving calmodulin¹⁰⁴; this is also involved in the regulated trafficking of AQP2.^{105,106} A critical role of the V_2R for urinary concentration has been demonstrated in two mouse models of X-linked nephrogenic diabetes insipidus (XNDI). Upon constitutive deletion,¹⁰⁷ male mutant mice ($V_2R^{-/-}$) die within 7 days after birth, with 3-day-old mice displaying severe hypernatremia,

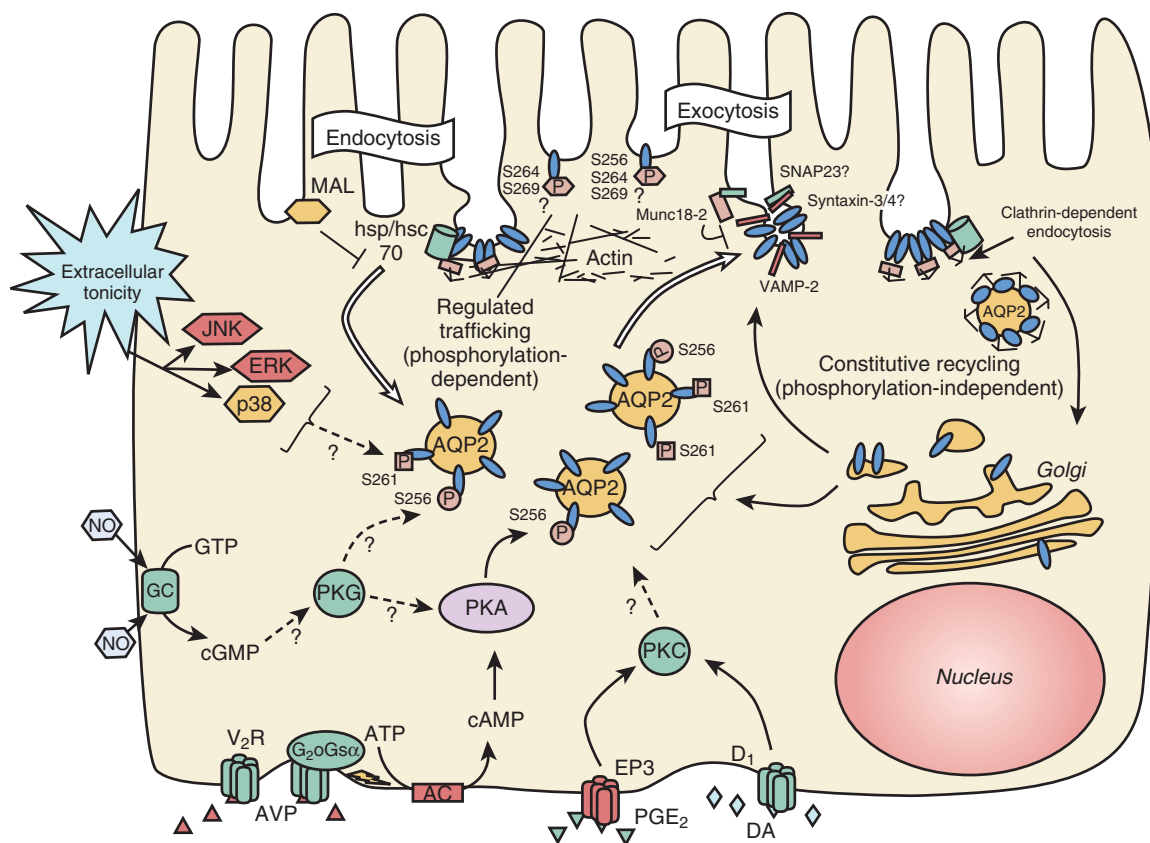


Fig. 10.9 Key events that contribute to the regulation of aquaporin-2 (AQP2) trafficking. The canonical pathway involves interaction of vasopressin with the type 2 receptor (V_2R) on the basolateral surface of the principal cell. This increases cyclic adenosine monophosphate (cAMP) formation after $G_{\alpha s}$ stimulation of adenylyl cyclase (AC). Phosphorylation of AQP2 occurs initially on residue S256, via protein kinase A (PKA) activation. After vasopressin stimulation, residue S261 on AQP2 is dephosphorylated, and residue S264 and S269 phosphorylation is increased. During exocytosis AQP2 interacts with soluble *N*-ethylmaleimide-sensitive factor attachment protein receptor (SNARE) proteins and their regulatory proteins such as Munc18-2, and these interactions may be regulated by phosphorylation. At the cell surface, phosphorylated AQP2 is present in endocytosis-resistant domains, and its interaction with heat shock protein/heat shock cognate 70 (hsp/hsc70), which is required for clathrin-mediated endocytosis, is inhibited. The myeloid and lymphocyte protein (MAL) also is involved in AQP2 endocytosis by an as-yet-unknown mechanism. Endocytosis of AQP2 is also facilitated by protein kinase C (PKC) activation (but possibly not by direct phosphorylation of AQP2), as well as by activation of dopamine (DA, D_1), prostaglandin E_2 (PGE_2), and PGE_2 receptor type 3 (EP3). However, constitutive exocytosis of AQP2 occurs without vasopressin stimulation and does not require AQP2 phosphorylation on residue S256. Accumulation of AQP2 at the plasma membrane is increased by inhibiting clathrin-mediated endocytosis. AQP2 phosphorylation can also be increased by stimulating the cyclic guanosine monophosphate/protein kinase G (cGMP/PKG) pathways using, for example, nitric oxide (NO). Extracellular hypertonicity activates the mitogen-activated protein (MAP) kinase pathway, and c-Jun N-terminal kinase (JNK), extracellular signal-regulated kinase (ERK), and p38 MAP kinase activities are all required for AQP2 surface accumulation after acute hypertonic shock. Finally, AQP2 trafficking involves the actin cytoskeleton, and actin depolymerization results in cell surface accumulation of AQP2 without the need for vasopressin stimulation. ATP, Adenosine triphosphate; GC, guanylyl cyclase; GTP, guanosine triphosphate; SNAP23, synaptosomal-associated protein 23; VAMP-2, vesicle-associated membrane protein 2.

drastically increased serum Na^+ and Cl^- levels, and significantly lower urine osmolality. In mice with conditional deletion of the V_2R ,^{108,109} adult mice display all of the characteristic symptoms of XNDI,^{110,111} including polyuria, polydipsia, and resistance to the antidiuretic actions of vasopressin.

The function of the V_2R depends on interaction with GPCRs and β -arrestin. Upon vasopressin binding, the V_2R assumes an active configuration and the bound heterotrimeric G protein, Gs, dissociates into $\text{Gs}\alpha$ and $\text{Gs}\beta\gamma$ subunits.¹⁰³ This G protein is localized on the basolateral plasma membrane of the thick ascending limb of Henle, distal convoluted tubule, and collecting duct principal cells.^{112,113} AC is stimulated by activated $\text{Gs}\alpha$, and cAMP levels are increased. The predominant AC isoform in the kidney is AC-6,¹¹⁴ and knockout mice lacking the AC-6 isoform have significant nephrogenic diabetes insipidus (NDI).^{115,116} After vasopressin binding,¹¹⁷ the V_2R is internalized, delivered to, and degraded in lysosomes, thus terminating the response. Many accessory proteins are involved in V_2R downregulation, including inhibitory G_i proteins,^{103,118,119} proteins involved in clathrin-mediated endocytosis,^{120,121} and proteins of the so-called retromer complex.^{122,123} Destruction of cAMP by cytosolic phosphodiesterases is also associated with limiting V_2R responses,¹²⁴ but cAMP levels in vasopressin target cells remain elevated for a considerable time after stimulation, and the V_2R continues to signal from endosomes after internalization.¹²²

A critical step in V_2R internalization is the binding of β -arrestin to the V_2R ,¹²⁵ which is triggered by phosphorylation of the V_2R by kinases, including G-protein-coupled receptor kinases (GRKs).¹²⁶ Following β -arrestin-dependent ubiquitination of the V_2R ,¹²⁷ arrestin-receptor complexes recruit the clathrin adaptor protein AP-2,¹¹⁹ and the complex is then internalized via clathrin-mediated endocytosis.^{120,128,129} Arrestins also uncouple GPCRs from GPCRs, producing a desensitized receptor.¹³⁰ Restoration of prestimulation levels of V_2R at the cell surface requires several hours.^{131–133} The majority of the V_2R that is internalized with vasopressin enters a lysosomal degradation compartment.^{127,134,135} Delivery of both the ligand and receptor to lysosomes may be required to terminate the physiologic response to vasopressin.¹³⁶ Restoration of prestimulation levels of the V_2R at the cell surface partly requires new protein synthesis.¹³⁴

VASOPRESSIN-REGULATED WATER TRANSPORT

COLLECTING DUCT WATER ABSORPTION AND OSMOTIC EQUILIBRATION

The urinary concentrating mechanism is dependent on two independent processes: (1) generation of a hypertonic medullary interstitium by concentration of NaCl and urea via countercurrent processes, and (2) osmotic equilibration of the tubule fluid within the medullary collecting ducts with the hypertonic medullary interstitium. As discussed, vasopressin is essential for determining the degree of water excretion because it increases NaCl reabsorption via the thick ascending limb and thus the hypertonicity of the medullary interstitium and it regulates collecting duct water permeability. When circulating vasopressin levels are low the water permeability of the collecting ducts is also extremely low; relatively little

water is reabsorbed from the tubule fluid and large volumes of hypotonic urine are produced. In contrast, high circulating levels of vasopressin increase the permeability of the apical membrane of the thick ascending limb to NaCl , leading to an increase in the osmolality of the peritubular interstitium (due to countercurrent multiplication) and increasing the water permeability of the collecting ducts to very high levels. Combined, this results in water being rapidly reabsorbed from the cortical and outer medullary portions of the collecting duct system via AQP water channels,^{137–139} resulting in the production of a small volume of hypertonic urine, with osmolality approaching that of the inner medullary interstitium.

The late distal tubule (the late distal convoluted tubule, the connecting tubule, and the initial collecting tubule) is the earliest site along the renal tubule where water absorption increases during antidiuresis (Fig. 10.10).¹⁴⁰ Although the

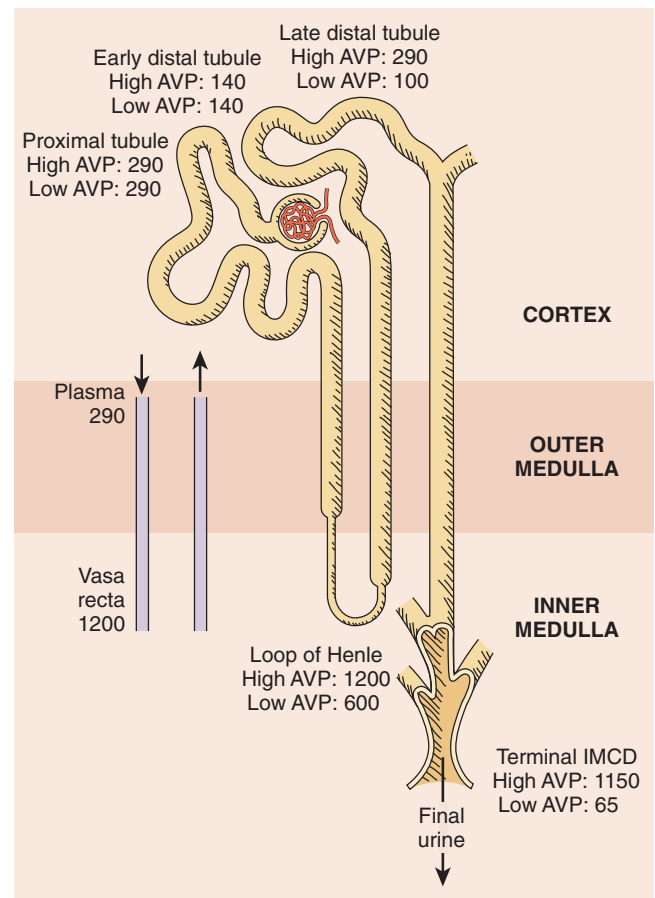


Fig. 10.10 Typical osmolalities (in mOsm/kg H_2O) found in various vascular (left) and renal tubule (right) sites in rat kidneys. Fluid in the proximal tubule is always isosmotic with plasma (290 mOsm/kg H_2O). Fluid emerging from the loop of Henle (entering the early distal tubule) is always hypotonic. Osmolality in the late distal tubule increases to plasma level only during antidiuresis. Final urine is hypertonic when the circulating vasopressin level is high, and hypotonic when the vasopressin level is low. A high osmolality is always maintained in the loop of Henle and vasa recta. During antidiuresis, osmolalities in all inner medullary structures are nearly equal. Osmolalities are somewhat attenuated in the loop and vasa recta during water diuresis (not shown). Based on micropuncture studies; see text. AVP, Vasopressin; IMCD, inner medullary collecting duct.

distal convoluted tubule does not express any water channels, it does express the V_2R and vasopressin regulates NaCl transport in this segment via increasing activity of the Na-Cl cotransport protein, NCC.^{141,142} In contrast, the connecting tubule and the cortical collecting duct express the V_2R and the vasopressin-regulated water channel AQP2.¹⁴³ Thus, it is likely that the connecting tubule and the cortical collecting duct segments are the earliest sites of distal tubular osmotic equilibration.

The volume of water absorption in the connecting segment and initial collecting tubule required to raise tubule fluid to isotonicity is considerably greater than the additional amount required to concentrate the urine above the osmolality of plasma in the medullary portion of the collecting duct system.³ Consequently, during antidiuresis, most of the water reabsorbed from the collecting duct system enters the cortical labyrinth, where the effective blood flow is high enough to return the reabsorbed water to the general circulation without diluting the interstitium. In contrast, if such a large volume of water was reabsorbed along the medullary collecting ducts, it would have a significant dilution effect on the medullary interstitium and thus impair concentrating ability).^{144,145}

During water diuresis, a modest corticomedullary osmolality gradient persists,^{146,147} and the water permeability of the

collecting ducts is low but not zero.^{148,149} Consequently, some water is reabsorbed by the collecting ducts during water diuresis, driven by the small transepithelial osmolality gradient. The majority of this water reabsorption occurs in the terminal inner medullary collecting ducts, where the transepithelial osmolality gradient is highest. In fact, more water is absorbed from the terminal inner medullary collecting ducts during water diuresis than during antidiuresis, owing to a much greater transepithelial osmolality gradient.^{144,145,150}

AQUAPORIN-2: THE VASOPRESSIN-SENSITIVE COLLECTING DUCT WATER CHANNEL

The first water channel, AQP1 was identified in 1991 by Peter Agre and his associates.^{151–154} AQP1 is expressed in proximal tubules and thin descending limbs of long-loop nephrons^{10,155,156} but not short-loop nephrons.¹⁷ AQP2, cloned in 1993, is the vasopressin-regulated water channel in kidney collecting duct principal cells.¹⁵⁷ Vasopressin stimulation of the collecting duct results in the accumulation of AQP2 on the plasma membrane of principal cells (Fig. 10.11). This involves the recycling of AQP2 between intracellular vesicles and the cell surface.^{137,158–164} However, aquaporin-3 (AQP3) and aquaporin-4 (AQP4), both of which are found in the

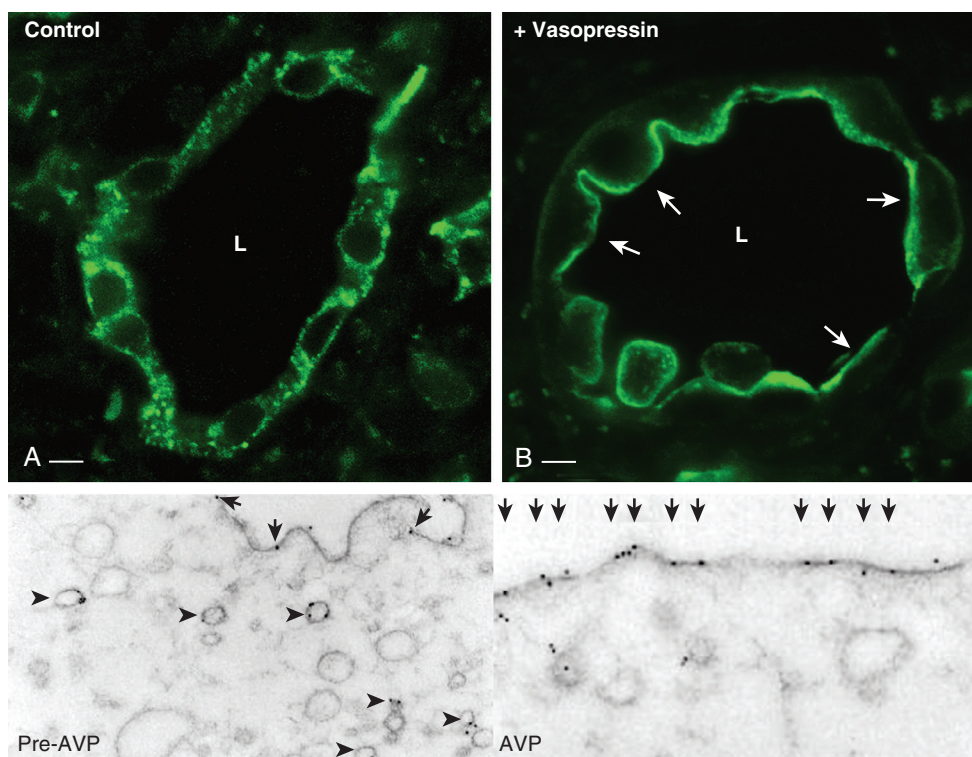


Fig. 10.11 Increased plasma membrane expression of AQP2 in principal cells of AVP-deficient Brattleboro rat kidney inner medullary collecting duct injected with vasopressin for 15 minutes. Kidneys were then fixed, sectioned and immunostained using anti-AQP2 antibodies. Under control conditions (A), AQP2 has a cytosolic distribution in principal cells. After perfusion with AVP (B), AQP2 shows an increased apical localization in principal cells (arrows). A weaker basolateral localization of AQP2 in principal cells is also visible in this section. The lower two panels show the effect of AVP on AQP2 distribution by immunogold electron microscopy. Tubules were perfused with 4 nM DDAVP for 60 minutes. The *left panel* (pre-AVP) shows the apical region of a principal cell, with gold particles (detecting AQP2) distributed on cytoplasmic vesicles, as well as a few on the apical plasma membrane (arrows). After AVP treatment, the number of gold particles on the apical plasma membrane is greatly increased (arrows), and the number of labeled cytoplasmic vesicles (arrowheads) is decreased. L, Tubule lumen. Scale bar, 5 μ m. (Lower panels adapted from Nielsen S, Chou CL, Marples D, et al. Vasopressin increases water permeability of kidney collecting duct by inducing translocation of aquaporin-CD water channels to plasma membrane. *Proc Natl Acad Sci U S A*. 1995;92:1013–1017.)

basolateral membrane of principal cells,^{165,166} are also regulated at the expression, and possibly functional level, by vasopressin and/or dehydration.^{166–169}

OVERVIEW OF VASOPRESSIN-REGULATED AQP2 TRAFFICKING IN COLLECTING DUCT PRINCIPAL CELLS

The vasopressin-induced change from a low-to-high permeability state of collecting duct principal cells, and vice versa, involves the reversible redistribution of AQP2 from cytoplasmic vesicles to the apical plasma membrane. Early freeze-fracture electron microscopy studies using amphibian urinary bladder and skin suggested that clusters of water channels are located on intracellular vesicles that fuse with the apical plasma membrane upon vasopressin stimulation. The water channels are internalized back into the cell by endocytosis after vasopressin washout.^{170–174} Antibodies against AQP2 demonstrated that it is located in the apical plasma membrane of collecting duct principal cells, as well as in intracellular vesicles.^{157,175,176} In vitro and in vivo studies correlated the vasopressin-stimulated increase in collecting duct water permeability and urinary concentration with relocalization of AQP2 from intracellular vesicles to the plasma membrane of principal cells (see Fig. 10.11).^{176–179} This relocation was reversible upon vasopressin washout and in animals either infused with a V₂R antagonist or subjected to water loading to reduce circulating vasopressin levels.^{180–182} One unexpected observation from initial studies was that significant amounts of AQP2 were present on principal cell basolateral membranes in some kidney regions, and that this staining tended to increase after vasopressin treatment. Recent studies have suggested that basolateral AQP2 is not only a potential pathway for water transport across the basolateral membrane, but may also have a role in cell migration and tubulogenesis due to interaction with β 1 integrin.^{183,184}

Some of the internalized AQP2 that accumulates in endosomes after vasopressin withdrawal follows a complex intracellular pathway before reinsertion into the plasma membrane.^{164,185–187} Unlike the V₂R,¹³⁴ de novo protein synthesis is not required for sequential responses to vasopressin stimulation.¹⁸⁸ A significant amount of AQP2 also accumulates in multivesicular bodies (MVBs).^{180,189} This pool of AQP2 can then be directed to lysosomes for degradation, be transferred to a recycling compartment, or be directly transported to the cell surface via transport vesicles that derive from the MVBs. The fate of internalized AQP2 seems to be at least in part regulated by ubiquitylation.^{190,193} Under certain conditions AQP2 can also be degraded in autophagosomes.^{194,195}

Some of the MVBs can fuse with the apical membrane of principal cells and release small nanovesicles known as exosomes into the tubule lumen. These exosomes contain a variety of different proteins,¹⁹⁶ including AQP2 on their limiting membranes,^{197,198} in addition to AQP2 mRNA and many other mRNAs and microRNAs within their lumen.^{199,200} AQP2 protein can be detected in urine, and the amount increases in conditions of antidiuresis, when more AQP2 is present in the apical membrane of principal cells. The physiologic relevance of this urinary excretion of AQP2 remains unknown, but the amount of exosomal AQP2 can be increased by vasopressin and urinary alkalization,²⁰¹ and a role in cell-cell communication has been proposed.²⁰⁰

BASOLATERAL AQUAPORINS IN PRINCIPAL CELLS

The presence of AQP3 and/or AQP4 renders the basolateral plasma membranes of collecting duct principal cells constitutively permeable to water.^{202,203} AQP3 expression is predominant in the cortex and decreases toward the inner medulla, with the reverse pattern for AQP4 (see Fig. 10.12), which is most abundant in the inner medulla.^{166,203} The abundances of AQP3 and AQP4 can be increased by the long-term action of vasopressin.^{202–204} AQP2 is also localized in the basolateral plasma membrane of these cells in some regions of the collecting duct.^{175,177,205–208} Basolateral expression of AQP2 is greatly increased by vasopressin^{209,210} or long-term (6 days) aldosterone.^{207,211,212} A proportion of basolateral AQP2 likely represents a transient step in an indirect apical targeting pathway for the AQP2 protein.^{213,214}

AQUAPORIN KNOCKOUT MICE

The physiological roles of various AQPs in the urinary concentrating mechanism have been uncovered by the use of various genetically modified mouse models.

AQUAPORIN-1 KNOCKOUT MICE

AQP1 knockout mice have increased urine volume and reduced urinary osmolality that does not increase in response to water deprivation.²¹⁵ Proximal tubule fluid absorption is markedly impaired in AQP1 knockout mice, but distal delivery of water and NaCl is not impaired due to a reduction in glomerular filtration rate via the tubular–glomerular feedback mechanism.²¹⁶ The osmotic water permeability of isolated perfused thin descending limbs from AQP1 knockout mice was markedly reduced compared with control animals.²¹⁷ As rapid water absorption from long-loop thin descending limbs is essential for countercurrent multiplication processes in the outer medulla, the reduced water reabsorption is one factor responsible for the concentrating defect in AQP1 knockout mice. Descending vasa recta, a second renal medullary site of AQP1 expression, also displayed a marked reduction in osmotic water permeability in AQP1 knockout mice,^{77,78} and thus countercurrent exchange processes are also likely to be impaired in AQP1 knockout mice. The results from studies in these mice show that AQP1 in the renal medulla is essential for the urine-concentrating mechanism.

AQUAPORIN-2 KNOCKOUT MICE

A number of different genetic models have been generated to assess the role of AQP2 in the urinary concentrating mechanism, including inducible and nephron-specific models of AQP2 deletion, models where essential phosphorylation sites in AQP2 are modified, and models of autosomal dominant NDI.^{192,218–226} The major phenotype in these models is severe polyuria; however, with free-access to water, plasma concentrations of electrolytes, urea, and creatinine are not different in knockout mice compared to controls. In contrast, a mouse model with connecting tubule-specific AQP2 deletion²²⁷ has indicated a role of the connecting tubule in regulating body water balance under basal conditions, but not for maximal concentration of the urine during antidiuresis. Taken together, these mouse models confirm that AQP2 is responsible for the

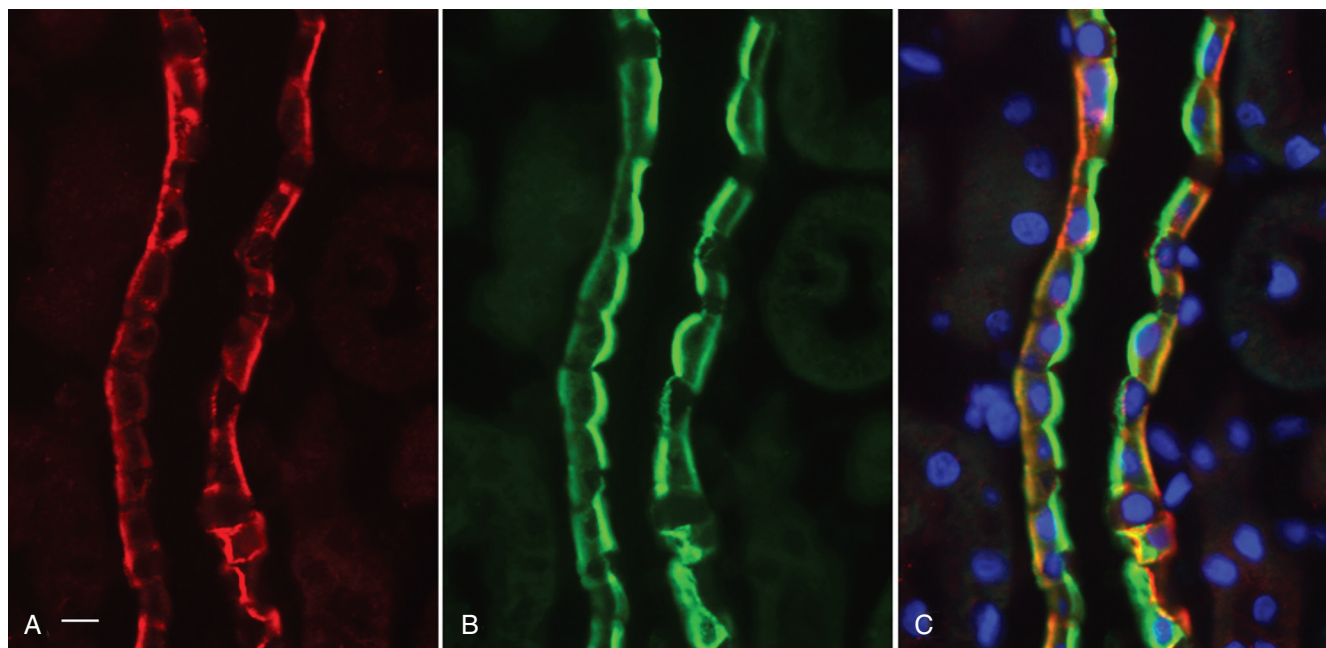


Fig. 10.12 Localization of aquaporins in the outer medullary collecting duct (outer stripe) of rat kidney. Images show sections immunostained in (A) for aquaporin-4 (AQP4) (red) and (B) for aquaporin-2 (AQP2) (green). The merged image in (C) shows that AQP2 is largely apical in this region, but both AQP2 and AQP4 are present on basolateral membranes. Intercalated cells are not stained with either antibody and appear as darker gaps among the other cells. In (C), nuclei are stained with 4',6-diamidino-2-phenylindole (DAPI). Scale bar, 10 μ m.

majority of transcellular water reabsorption in the connecting tubule and collecting duct system.

AQUAPORIN-3 AND AQUAPORIN-4 KNOCKOUT MICE

The osmotic water permeability of the basolateral membrane of cortical collecting duct cells from AQP3 knockout mice is reduced by greater than threefold compared with wild type control mice.²²⁸ Consequently, AQP3 knockout mice are markedly polyuric (10-fold greater daily urine volume than controls), but they can slightly increase their urine osmolality after either water deprivation or vasopressin treatment.³ AQP4 knockout mice have a fourfold decrease in inner medullary collecting duct osmotic water permeability, indicating that AQP4 is responsible for the majority of water movement across the basolateral membrane in this segment.^{229,230} Despite this reduced inner medullary collecting duct water permeability, AQP4 knockout mice have no difference in urine osmolality. However, after 36 hours of water deprivation, AQP4 knockout mice have a significantly reduced maximal urine osmolality that cannot be further increased by AVP administration. This modest decrease in urinary concentrating ability in AQP4 knockout mice, compared with the profound concentrating defect in AQP3 knockout mice, is likely due to the normal distribution of water transport along the collecting duct,³ with much greater osmotic reabsorption of water in the cortical portion of the collecting duct system (where AQP3 is predominant), than in the medullary collecting ducts (where AQP4 is the predominant basolateral water channel).

MECHANISMS OF AQUAPORIN-2 TRAFFICKING

A wealth of information regarding the regulated trafficking, function, structure, and water transport capacity of AQP2

has been generated by the use of various in vitro or ex vivo experimental systems.^{154,178,188,231–250} In the following sections, we discuss the various mechanisms of AQP2 trafficking that are continually evolving in parallel with new discoveries related to the targeting and trafficking of membrane proteins in general.

AQUAPORIN-2 RECYCLING

Clathrin-coated pits are critical for the internalization of both AQP2 and the V_2R ,^{119,120,128,251,252} with inhibition of clathrin-mediated endocytosis causing AQP2 plasma membrane accumulation (Fig. 10.13).^{159,185,186,251,253–256} Although caveolae have been proposed as an alternative endocytotic pathway for AQP2 in cultured cells,²⁵⁷ caveolae and caveolin are not present on the apical pole of principal cells in vivo.^{258,259} After internalization, AQP2 enters a subapical recycling compartment distinct from organelles such as the Golgi, the trans-Golgi network (TGN), and lysosomes,^{188,260,261} but it likely recycles via the classic endosomal recycling compartments^{186,262,263} with the vacuolar protein sorting-associated protein 35 (Vps35) playing an important role.²⁶⁴

ROLE OF THE CYTOSKELETON IN AQUAPORIN-2 TRAFFICKING

The cytoskeleton and soluble N-ethylmaleimide-sensitive factor attachment protein receptor (SNARE) complex play important roles in multiple aspects of vesicle trafficking, including exocytosis, endocytosis, and vesicle docking and fusion.^{265–273} Thus, it is no surprise that they are important for AQP2 trafficking.

Actin associates directly with AQP2^{274–276} or AQP2-containing vesicles²⁷⁷ and upon vasopressin-mediated depolymerization, AQP2 accumulates in the plasma membrane.^{278–281} Apical fluid shear stress also depolymerizes the apical actin cytoskeleton and causes AQP2 membrane accumulation.^{282,283} A

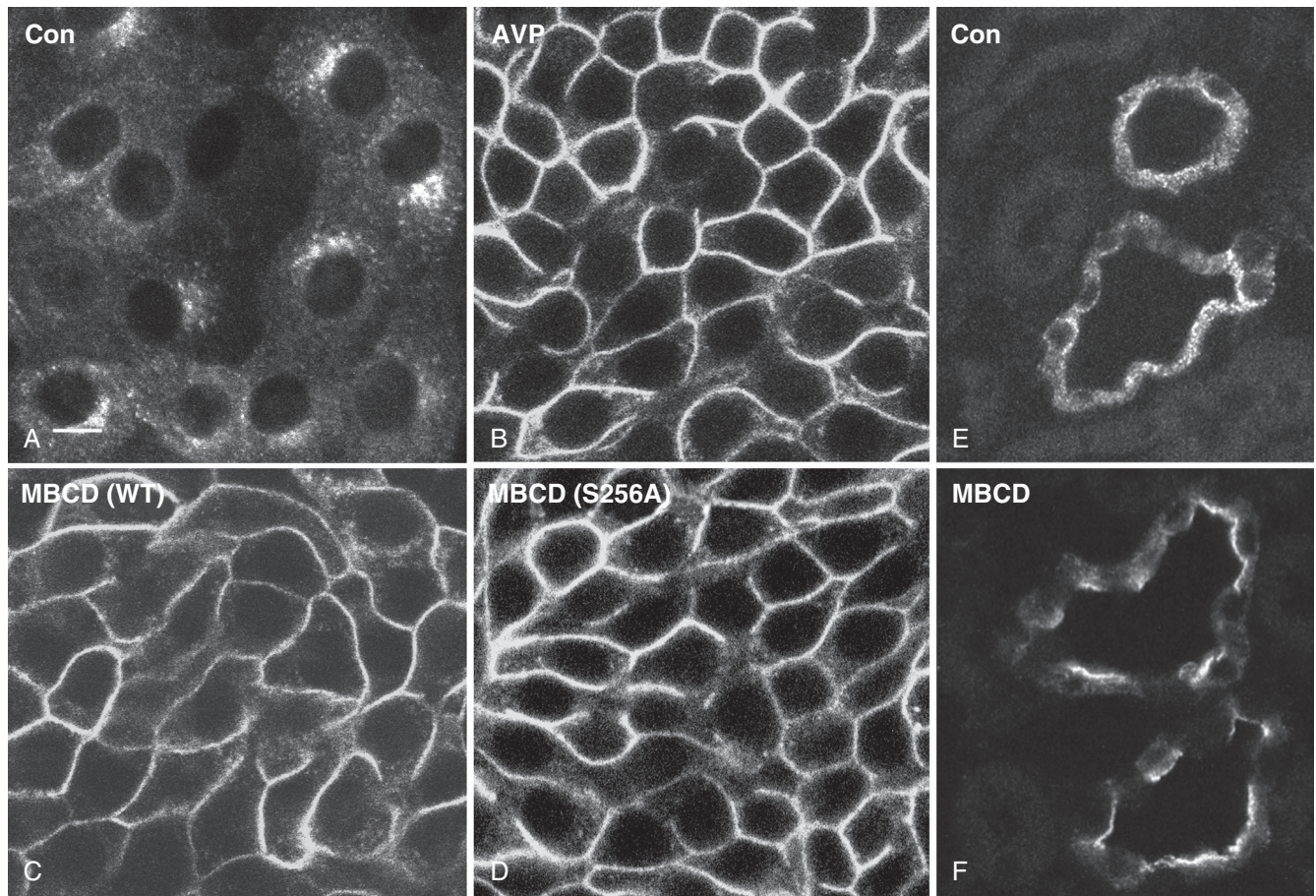


Fig. 10.13 Methyl- β -cyclodextrin (MBCD) stimulates aquaporin-2 (AQP2) membrane accumulation in LLC-PK₁ cells (A) to (D) and collecting duct principal cells in situ (E) and (F). Immunofluorescence staining for AQP2 in LLC-PK₁ cells expressing wild type AQP2 (A) to (C) or a mutant in which the S256 residue has been replaced by alanine (S256A) (D). Under baseline conditions, wild type AQP2 is located mainly on intracellular vesicles, often concentrated in the perinuclear region of the cell (A). After vasopressin (AVP) treatment, wild type AQP2 relocates to the plasma membrane (B). When endocytosis is inhibited by application of the cholesterol-depleting drug MBCD, both wild type and S256A AQP2 accumulate at the cell surface in the absence of AVP (C) and (D). This result shows that both wild type AQP2 and S256A AQP2 are constitutively recycling between intracellular vesicles and the plasma membrane, and that inhibiting endocytosis with MBCD is sufficient to cause membrane accumulation, even in the absence of S256 phosphorylation of AQP2. In collecting duct principal cells (inner stripe of outer medulla) in situ, AQP2 is located on vesicles scattered throughout the cytoplasm after perfusion of intact kidneys in vitro (E). However, after perfusion of kidneys for 60 minutes with 5 mmol/L MBCD, increased apical plasma membrane expression of AQP2 is seen (F). This finding indicates that AQP2 is constitutively recycling through the apical plasma membrane in principal cells in situ, and that membrane accumulation can be induced by blocking endocytosis (with MBCD) even in the absence of vasopressin. *Con*, Control; *WT*, wild type.

role for A-kinase anchoring protein 220 (AKAP220) and Rho GTPases in modulating the actin effects on AQP2 have been proposed.^{280,284–288} AQP2 also complexes with various other actin-associated proteins including myosins,^{277,281,289–291} Rab proteins,^{262,292} members of the ERM (ezrin-radixin-moesin) family,^{293,294} and the signal-induced proliferation-associated gene 1 (SPA-1).²⁸⁸ Although the mechanisms behind actin depolymerization and AQP2 trafficking are not clear (see Fig. 10.14), only vasopressin induces significant actin depolymerization in cells expressing AQP2,²⁸³ suggesting a novel mechanism of protein trafficking in which the channel protein itself critically regulates local actin reorganization to initiate its movement.²⁷⁵ The integrin-linked kinase (ILK) is also important in orchestrating cytoskeletal organization during AQP2 recycling and entry into the exocytotic pathway.^{295,296}

Dynein and dynactin, a protein complex linking microtubules and vesicles, are associated with AQP2-bearing vesicles²⁹⁷

and depolymerization of microtubules partially inhibits vasopressin-induced water permeability in target epithelia^{298–300} and apical localization of AQP2.^{176,249,301,302} A role of microtubules in the basolateral to apical transcytosis of AQP2 has also been suggested.²¹⁴ Together, the data on microtubules indicate that they are predominantly responsible for long-range trafficking of AQP2 vesicles toward the plasma membrane and localization of AQP2 inside the cell after internalization, but that the final steps of vesicle approach and fusion are microtubule independent.³⁰³

A variety of SNARE proteins are associated with AQP2-containing vesicles or colocalize with AQP2 in collecting duct cells, including VAMP-2 (vesicle-associated membrane protein 2, synaptobrevin-2), VAMP-3 (cellubrevin), VAMP-8, SNAP23 (synaptosomal-associated protein 23), the ATPase Hrs-2, syntaxin 3, syntaxin 4, syntaxin 7, syntaxin 12, and syntaxin 13.^{277,304–310} Of these, VAMP-2, VAMP-3, syntaxin 3,

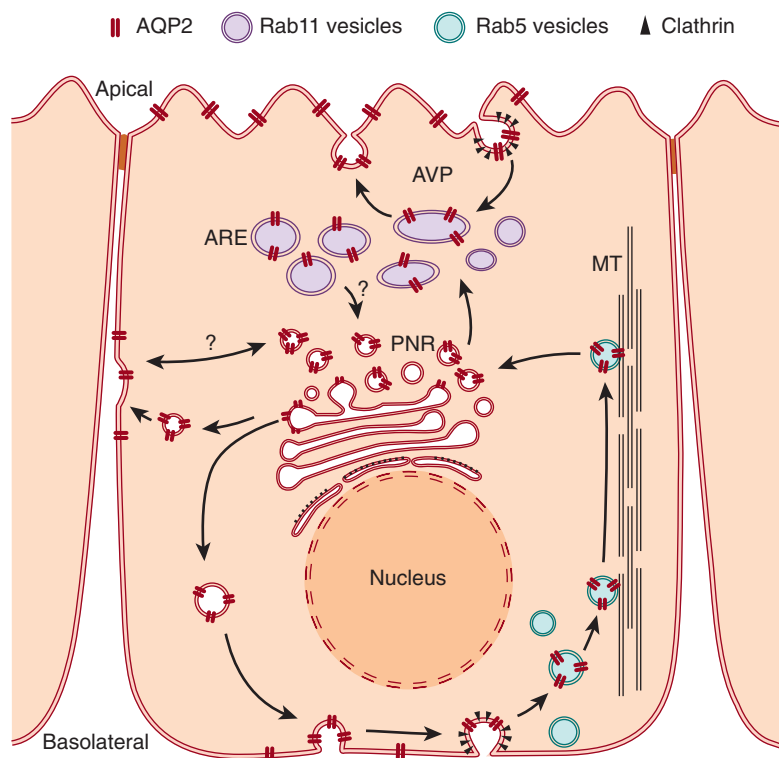


Fig. 10.14 Aquaporin-2 (AQP2) follows a transcytotic pathway before apical membrane delivery. From vesicles in the perinuclear region (PNR), probably originating from the trans-Golgi network, AQP2 can be delivered to the basolateral plasma membrane before reaching the apical surface of epithelial cells. From there, it is retrieved by clathrin-mediated endocytosis into Rab5-positive endosomes (green), which move in a microtubule (MT)-dependent manner to the PNR and ultimately to Rab11-positive apical recycling endosomes (AREs; purple). These Rab11-positive vesicles are involved in recycling AQP2 constitutively to and from the apical plasma membrane. The endocytotic branch of this recycling pathway is inhibited by the methyl- β -cyclodextrin treatment shown in Fig. 10.13, resulting in cell surface accumulation of AQP2. The physiologic stimulus, vasopressin (AVP), increases apical AQP2 expression in two ways. It increases exocytosis from the Rab11 compartment and also inhibits clathrin-mediated endocytosis of AQP2 from the apical plasma membrane. The delivery of AQP2 to the basolateral membrane of collecting duct principal cells may be important for collecting duct tubulogenesis,¹⁸³ whereas apical AQP2 is necessary for urine concentration. From Yui N, Lu HA, Chen Y, et al. Basolateral targeting and microtubule-dependent transcytosis of the aquaporin-2 water channel. *Am J Physiol Cell Physiol.* 2013;304:C38–C48.

and SNAP23 are the ones that have been functionally shown to be important for AQP2 trafficking.^{306,309,311} Interaction of AQP2 and the SNARE complex may be mediated by the protein snapin³¹² and/or by the angiotensin-converting enzyme 2 homolog collectrin, which has been implicated in salt-sensitive hypertension.³¹³

ESSENTIAL ROLE OF AQUAPORIN-2 PHOSPHORYLATION

The rise in intracellular cAMP following V₂R stimulation is important for modulating the abundance of AQP2 by modulation of AQP2 gene transcription.^{242,314} cAMP also plays a role in AQP2 trafficking by affecting the phosphorylation status of AQP2,^{262,315–317} with phosphatase inhibitors increasing cell surface accumulation of AQP2.^{318,319} However, V₂R-mediated increases in cAMP are not absolutely necessary for receptor-mediated AQP2 membrane targeting and alternative pathways to increase AQP2 membrane targeting exist.^{320–323}

AQP2 contains several phosphorylation sites for protein kinases,^{138,324–326} several of which are important for AQP2 trafficking alongside AQP2 ubiquitylation.^{190–193} Whether any of the phosphorylation sites are important for modulation of AQP2 unit water permeability is controversial.^{327–329} Early

work focused on the involvement of S256 phosphorylation in AQP2 trafficking, with the current consensus being that S256 phosphorylation is necessary for vasopressin-induced cell-surface accumulation of AQP2^{222,330–334} (Fig. 10.15). The importance of this site is highlighted by a mutation (S254L), which destroys the PKA phosphorylation site at S256 resulting in NDI in humans.^{334,335} The roles of S261, S264, and S269 (threonine in humans) are slowly being uncovered.³²⁸ All three phosphorylated forms are localized to some degree in the plasma membrane in vivo.^{169,189,336,337} Vasopressin decreases the abundance of pS261, whereas AMP-activated kinase (AMPK) activation increases the levels.³³⁸ However, this site alone is not required for AQP2 trafficking.^{339,340} Interestingly, activation of AMPK with metformin increases AQP2 phosphorylation in general,³⁴¹ whereas levels of phosphorylation are significantly attenuated under acidic conditions.³⁴² The pS269 form of AQP2 is exclusively detected in the apical plasma membrane, and a regulatory role of this phosphorylation site directly in the plasma membrane for inhibiting AQP2 endocytosis has been shown.*

*References 169, 189, 191, 336, 337, and 343.

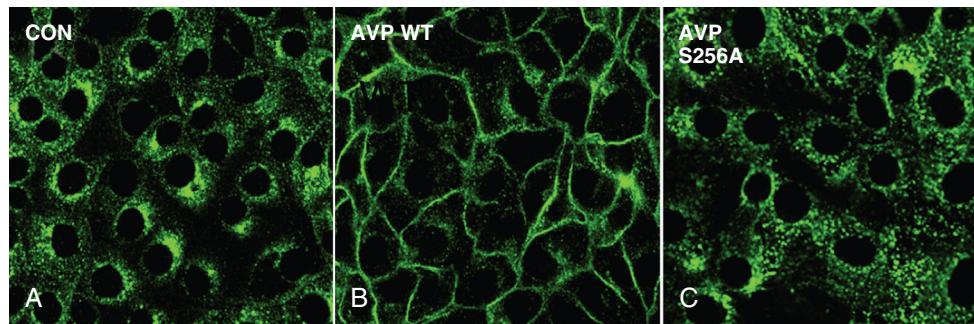


Fig. 10.15 Immunofluorescence staining showing aquaporin-2 (AQP2) expressed in LLC-PK₁ cells. Under control (CON) conditions (A), AQP2 is located on perinuclear and more diffusely distributed intracellular vesicles, with very little plasma membrane staining. After vasopressin (AVP) treatment for 10 minutes, AQP2 accumulates on the plasma membrane of cells expressing wild type (WT) AQP2 (B) but remains mainly on intracellular vesicles after vasopressin treatment of cells expressing AQP2-S256A, a mutation that prevents protein kinase A-mediated phosphorylation of this critical amino acid (C).

ROLE OF PHOSPHORYLATION IN EXOCYTOSIS AND ENDOCYTOSIS OF AQUAPORIN-2

Although S256 phosphorylation is necessary for vasopressin-induced cell-surface accumulation of AQP2, the role that phosphorylation plays in AQP2 exocytosis is complex. An AQP2 S256A mutant accumulates on the plasma membrane upon inhibition of endocytosis (see Fig. 10.13),²⁵³ suggesting that the exocytotic pathway is intact under these conditions. Vasopressin also increases exocytosis of vesicles in AQP2-expressing cells whether or not AQP2 is phosphorylated at S256.³⁴⁴ Thus, although vasopressin-induced accumulation of AQP2 at the cell surface requires S256 phosphorylation, and AQP2 is present in “endocytosis-resistant” membrane domains after vasopressin treatment,^{253,345,346} exocytotic insertion of AQP2 into the plasma membrane is probably independent of this phosphorylation event. Furthermore, the regulated endocytosis of AQP2 may not be dependent on its phosphorylation state.³⁴⁷ For example, prostaglandin E₂ (PGE₂)³⁴⁸ can induce AQP2 internalization independent of S256 phosphorylation, but other studies indicate that the effects of PGE₂ on AQP2 and urine concentration depend on which PGE₂ receptor it acts upon.^{109,250,344,349–351}

Accumulating evidence suggests that phosphorylation-mediated interaction of AQP2 with other regulatory proteins is important for modulating cell-surface accumulation of AQP2. For example, AQP2 phosphorylation modifies its interaction with key proteins of the vesicle docking/fusion apparatus or endocytotic machinery, including heat shock cognate/heat shock protein 70 (hsc/hsp70)^{345,352} dynamin and clathrin,^{340,352} annexin 2,¹⁶⁹ the myelin and lymphocyte protein (MAL),³⁵³ or 14-3-3 ζ .³⁵⁴

VASOPRESSIN-REGULATED UREA TRANSPORT IN THE INNER MEDULLA

UREA TRANSPORTER PROTEINS

Urea plays a central role in the urinary concentrating mechanism. Urea importance has been appreciated since 1934, when Gamble and colleagues initially described “an economy of water in renal function referable to urea,”³⁵⁵ findings which were recently confirmed and advanced in UT-A1/A3 knockout mice³⁵⁶ (discussed later). Many studies

show that maximal urine-concentrating ability is decreased in protein-deprived or malnourished humans (and other mammals), and that urea infusion restores urine-concentrating ability (reviewed in Sands and Layton³⁵⁷). Urine-concentrating defects have been demonstrated in UT-A1/A3,³⁵⁸ UT-A2,³⁵⁹ UT-B,^{360–362} and UT-A2/UT-B knockout mice.³⁶³ Thus, an effect due to urea or UTs must be part of the mechanism by which the inner medulla concentrates urine.

Two urea transporter genes have been cloned in mammals: the UT-A (*Slc14A2*) gene encodes 6 protein and 9 cDNA isoforms (reviewed in Sands and Layton³⁵⁷), and the UT-B (*Slc14A1*) gene encodes 2 protein isoforms.³⁶⁴ The UT-A gene, which has been cloned from rodents and humans, has two promoter elements: one upstream of exon 1 and a second that is located within intron 12 and drives the transcription of UT-A2 and UT-A2b (see references 365–368; also reviewed in Sands and Layton³⁵⁷). UT-B, which is also the Kidd blood group antigen in humans, has been cloned from humans and rodents³⁶⁹ (also reviewed by Sands and Layton³⁵⁷).

UT-A promoter I contains a tonicity enhancer (TonE) element and hyperosmolality increases its activity.^{366,370} UT-A1 is expressed in the terminal inner medullary collecting duct and is detected in the apical plasma membrane.^{367,371,372} UT-A3 is also expressed in the terminal inner medullary collecting duct; it is primarily detected in the basolateral plasma membrane but has been detected in the apical plasma membrane.^{373–375} UT-A2 is expressed in thin descending limbs.^{8,371,372,376} UT-B is expressed in descending vasa recta and red blood cells (reviewed by Sands and Layton³⁵⁷) (Fig. 10.16).

Vasopressin increases the phosphorylation and the apical plasma membrane accumulation of UT-A1 and of UT-A3 in rat inner medullary collecting ducts.^{375,377} UT-A1 is phosphorylated by vasopressin at serines 486 and 499.^{325,378} Both phospho-S486-UT-A1 and phospho-S499-UT-A1 are expressed predominantly in the apical plasma membrane in vasopressin-treated rat inner medullary collecting ducts.^{379,380} The site in UT-A3 that is phosphorylated by vasopressin has not been determined, except that neither of the two PKA consensus sites is involved.³⁸¹ Vasopressin stimulates urea transport, UT-A1 phosphorylation, and apical plasma membrane accumulation through two cAMP-dependent pathways: PKA and Epac (exchange protein activated by cAMP).³⁸² Epac increases UT-A1 phosphorylation but not at either serine 486 or 499.³⁸⁰

UT-A1 is dephosphorylated by multiple phosphatases, including calyculin and calcineurin.³⁸³ 14-3-3 proteins bind

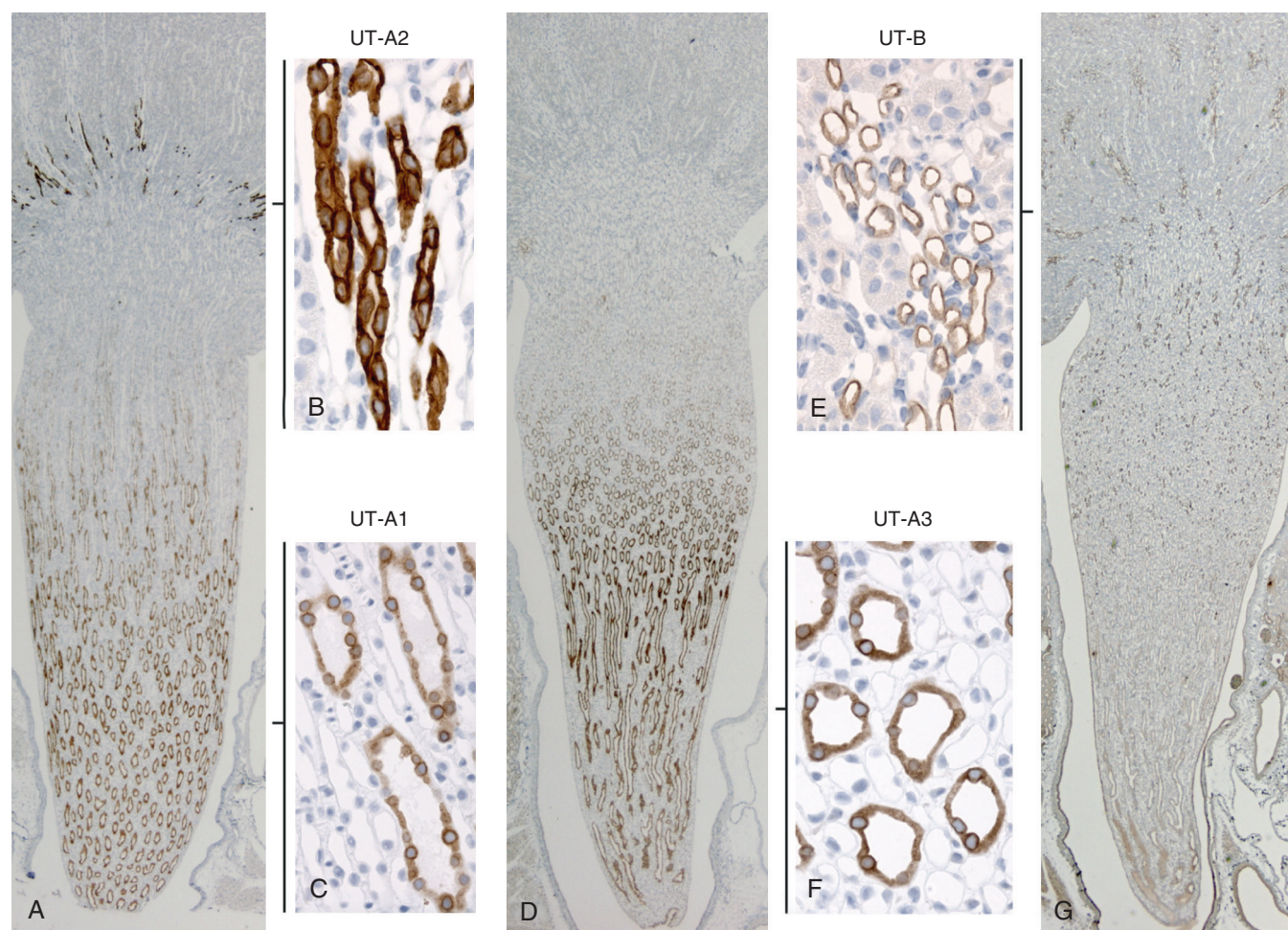


Fig. 10.16 Localization of urea transporters. UT-A1 is localized to the terminal portion of the inner medullary collecting duct, whereas UT-A2 is localized to the thin descending limbs of Henle loop in the inner stripe of outer medulla (A). Higher magnification shows that both UT-A2 (B) and UT-A1 (C) are predominantly intracellular. UT-A3 is localized to the terminal portion of the inner medullary collecting duct (D) and is both intracellular and in the basolateral membrane domains (F). UT-B is expressed in the descending vasa recta (G), where it is localized to the basolateral and apical regions (E). Adapted from Fenton RA, Knepper MA. Urea and renal function in the 21st century: insights from knockout mice. *J Am Soc Nephrol.* 2007;18: 679–688.

to phosphorylated serine or threonine residues and regulate protein function. UT-A1 and 14-3-3 γ bind, and PKA activation enhances this binding.³⁸⁴ 14-3-3 γ increases UT-A1 ubiquitination and degradation by interacting with the E3 ubiquitin ligase, MDM2, and decreases urea transport.³⁸⁴ Thus, UT-A1 phosphorylation is increased by PKA, and UT-A1 degradation is enhanced by subsequent binding to 14-3-3 γ , potentially providing a negative feedback mechanism to return UT-A1 function to its basal state following vasopressin stimulation.³⁸⁴ Although data showing these opposite effects of vasopressin/PKA are established, the physiologic significance remains to be determined.

Hyperosmolality increases urea permeability in rat terminal inner medullary collecting ducts, even in the absence of vasopressin,^{385–387} suggesting that it is an independent activator of urea transport. Hyperosmolality stimulates urea permeability via activation of PKC α and intracellular calcium,^{388–391} whereas vasopressin stimulates urea permeability via increases in cAMP.³⁹² Hyperosmolality increases the phosphorylation and the plasma membrane accumulation of both UT-A1 and UT-A3,^{375,377,393,394} similar to the effect of vasopressin. UT-A1 is phosphorylated by PKC α at serine 494.^{390,395–397}

Genetic knockout of PKC α in mice results in a urine-concentrating defect.^{395,398,399} PKC α knockout mice have reduced levels of UT-A1 protein abundance³⁹⁵ and UT-A1 sialylation.⁴⁰⁰ PKC α activation increases UT-A1 sialylation and UT-A1 accumulation in the apical plasma membrane, an effect mediated by Src kinase.⁴⁰⁰ PKC α also enhances UT-A3 sialylation, an effect mediated by ST6GalI.⁴⁰¹

Metformin, an AMPK activator, increases UT-A1 and AQP2 phosphorylation in inner medullary collecting duct suspensions, urea and water transport in perfused terminal inner medullary collecting ducts, and urine-concentrating ability in two rodent models of congenital NDI: tolvaptan-treated rats and V2R knockout mice.^{341,402} Thus, drugs that activate AMPK may be a future therapy for NDI.^{341,402,403}

UREA TRANSPORTER KNOCKOUT MICE

UT-A1/A3 KNOCKOUT MICE

A mouse model where the two inner medullary collecting duct UTs, UT-A1 and UT-A3, are deleted (*UT-A1/A3*^{-/-} mice) have a complete absence of phloretin-sensitive and vasopressin-regulated urea transport in the inner medullary collecting duct.^{358,404–406} These *UT-A1/A3*^{-/-} mice fed a normal or high protein diet have

a significantly greater fluid intake and urine flow, resulting in a decreased urine osmolality, compared with wild type mice.^{358,407} Under these dietary conditions, after an 18-hour water restriction, *UT-A1/A3*^{-/-} mice are unable to reduce their urine flow to levels below those observed under basal conditions, resulting in volume depletion and loss of body weight. In contrast, on a low-protein diet (4%), *UT-A1/A3*^{-/-} mice do not show a substantial degree of polyuria and can reduce their urine volume to a similar level as control mice after water restriction. On a low-protein diet, hepatic urea production is low and urea delivery to the inner medullary collecting duct is predicted to be low, thus rendering collecting duct urea transport largely immaterial to water balance. Thus, the concentrating defect in *UT-A1/A3*^{-/-} mice is due to a urea-dependent osmotic diuresis, results that are compatible with a model of urea handling proposed in the 1950s by Berliner and colleagues.⁸³

UT-A1/A3^{-/-} mice have also been exploited to study the “passive mechanism for urine concentration models” proposed in 1972 by Kokko and Rector and by Stephenson for concentration of Na⁺ and Cl⁻ in the inner medulla in the absence of active transport^{408,409} (see later). In these models, the passive electrochemical gradient that drives Na⁺ and Cl⁻ to exit from the thin ascending limb is indirectly dependent on rapid reabsorption of urea from the inner medullary collecting duct. However, despite a profound decrease in inner medulla urea accumulation in *UT-A1/A3*^{-/-} mice, three independent studies failed to demonstrate the predicted decline in Na⁺ and Cl⁻ concentrations in the inner medulla.^{355,358,405,406} Based on these results alone, the passive concentrating model in the form originally proposed does not appear to be the only mechanism by which NaCl is concentrated in the inner medulla. However, mathematical modeling analysis of these same data concluded that the results found in the *UT-A1/A3*^{-/-} mice are consistent with what one would predict for the passive mechanism.⁴¹⁰ Thus, the issue remains unresolved at present.

Another hypothesis regarding urea and the urinary concentrating mechanism was described over 80 years ago as “an economy of water in renal function referable to urea” and affectionately known as the Gamble phenomenon.³⁵⁵ Gamble described that (1) the water requirement for excretion of urea is less than for excretion of an osmotically equivalent amount of NaCl, and (2) less water is required for the excretion of urea and NaCl together than the water needed to excrete an osmotically equivalent amount of either urea or NaCl alone. In *UT-A1/A3*^{-/-} mice, both elements of the Gamble phenomenon were absent, indicating that inner medullary collecting duct UTs play an essential role.³⁵⁶ When wild type mice were given progressively increasing amounts of urea or NaCl in the diet, both substances induced osmotic diuresis, but at different excretion levels (6000 μ osmol/day for urea; 3500 μ osmol/day for NaCl). Mice were unable to increase urinary NaCl concentrations above 420 mM. Thus, the second component of the Gamble phenomenon derives from the fact that both urea and NaCl excretion are saturable, presumably resulting from an ability to exceed the respective reabsorptive capacity for urea and NaCl, rather than a specific interaction of urea transport and NaCl transport at an epithelial level.

A mouse lacking UT-A3 but expressing UT-A1 was created by transgenic restoration of UT-A1 into the *UT-A1/A3*^{-/-} knockout mouse in order to determine the effect of UT-A1 alone.⁴¹¹ Basal urea permeability in the inner medullary collecting duct of the UT-A1-only mouse was normal, but unlike wild type mice, vasopressin did not stimulate urea permeability

above basal levels.⁴¹¹ Surprisingly, urine-concentrating ability was restored to wild type levels in the UT-A1 only mice.⁴¹¹

UT-B AND UT-A2 KNOCKOUT MICE

UT-B knockout mice have a reduced urine-concentrating ability that is similar to humans lacking UT-B (reviewed by Fenton and Knepper⁴⁰⁶ and Klein et al.⁴¹²). In humans, UT-B is the Kidd blood group antigen, and people lacking the Kidd antigen are unable to concentrate their urine above 800 mOsm/kg H₂O, even following overnight water deprivation and exogenous vasopressin administration.⁴¹³

Mice lacking UT-A2 also have reduction in urine-concentrating ability (reviewed in Fenton and Knepper⁴⁰⁶ and Klein et al.⁴¹²). The urine-concentrating defect is thought to result from impairment of urea recycling (reviewed in Fenton and Knepper⁴⁰⁶ and Klein et al.⁴¹²). Because UT-B knockout may also interfere with urea recycling, a mouse lacking both UT-B and UT-A2 was generated.³⁶³ Unexpectedly, UT-A2 deletion appeared to partially correct the concentrating defect in mice lacking only UT-B.³⁶³ These results suggest that rather than playing a role in maintaining urea concentration during the normal steady state, UT-A2 may function to move urea during the acute transition from diuresis to antidiuresis.³⁶³

MICE LACKING ALL UREA TRANSPORTERS

Mice lacking all UTs have a 3.5-fold increase in urine output, produce dilute urine, and have reduced blood pressure.⁴¹⁴ The all-UT knockout mice do not increase urine osmolality or urea following water restriction, acute urea loading, or a high protein intake.⁴¹⁴ The all-UT knockout mice do not exhibit physiological abnormalities in extrarenal tissues.⁴¹⁴

Clinical Relevance

Urearetics

In recent years, urea transporter inhibitors have been developed as potential novel diuretics (reviewed in references 476 through 478). Dimethylthiourea (DMTU), a urea analog, inhibits UT-A1 and UT-B, results in a sustained and reversible reduction in urine osmolality, an increase in urine volume, and mild hypokalemia in rats.^{479,480} Other thiourea analogs are being investigated for selective inhibition of UT-A1 or UT-B.⁴⁷⁹ Another class of inhibitors, an indole thiazole or γ -sultambenzosulfonamide, is selective for UT-A and results in diuresis with more urea than salt excretion in rats, even when the rats were given dDAVP.⁴⁸¹ Another potential class of inhibitors are 2,7-distributed fluorenones, the most potent of which inhibited UT-A1 and UT-B with an IC₅₀ of 1 μ M.⁴⁸² A fourth class are thienoquinolins that inhibit both UT-A and UT-B, PU-14, results in a diuresis in rats.^{483,484} The thienoquinolin PU-48 results in a diuresis in both wild type and UT-B knockout mice, indicating that its effect was to inhibit UT-A, and inhibits urea permeability in perfused rat inner medullary collecting ducts.⁴⁸⁵ Because the diuresis induced by PU-48 did not change serum sodium, chloride, or potassium levels, it supports the hypothesis that an agent that targets UT-A1, which is expressed in the last portion of the inner medullary collecting duct, may have less risk for side effects, such as hypokalemia, than conventional diuretics that act in more proximal portions of the nephron.⁴⁸⁵

ACCUMULATION OF UREA IN RENAL INNER MEDULLA

Urea accumulation within the inner medulla is partly dependent on variable urea permeabilities along the collecting duct system (Fig. 10.17). Within the collecting duct system, only the terminal inner medullary collecting duct possesses high urea permeability,⁴¹⁵ which can be further increased by vasopressin.^{148,416,417} UT-A1 and UT-A3 UTs are localized to the apical and basolateral plasma membranes of the inner medullary collecting duct cells and are responsible for the high urea permeability of the terminal portion of the inner medullary collecting duct. The mechanisms of urea accumulation in the renal medulla are depicted in Fig. 10.18. Accumulation of urea is predominantly a result of passive urea reabsorption from the inner medullary collecting duct. Tubular fluid entering the collecting duct system in the renal cortex has a relatively low urea concentration. However, during antidiuresis, water is osmotically reabsorbed from the urea-impermeable parts of the collecting duct system in the cortex and outer medulla, causing a progressive increase in the luminal urea concentration along the connecting tubules, cortical collecting ducts, and outer medullary collecting ducts. Thus, when the tubule fluid reaches the highly urea permeable terminal inner medullary collecting duct (due to the presence of UTs), urea rapidly exits from the lumen to the inner medullary interstitium, where it is “trapped” by countercurrent urea exchange between descending and ascending flows in both the vasa recta and loops of Henle. Under steady-state conditions, and in the continued presence of vasopressin, urea nearly equilibrates across the inner medullary collecting duct epithelium and thus osmotically balances the urea in the collecting duct lumen, preventing possible instances of osmotic diuresis (Fig. 10.19).

The descending and ascending vasa recta are in close association with each other in the inner medulla, facilitating

countercurrent exchange of urea between the two structures.⁸³ In the ascending vasa recta, aided by the extremely high ($>40 \times 10^{-5}$ cm/sec) permeability to urea, the concentration of urea exiting the inner medulla is similar to the concentration of urea in the descending vasa recta.^{79,416} This minimizes the washout of urea from the inner medulla. However, countercurrent exchange cannot completely eliminate loss of urea from the inner medullary interstitium, because the volume flow rate of blood in the ascending vasa recta exceeds that in the descending vasa recta.⁴¹⁸ During antidiuresis, water is added to the vasa recta from both inner medullary collecting ducts and descending limbs, resulting in a higher volume flow rate and an increased mass flow rate of urea. This ensures that the inner medullary vasculature continually removes urea from the inner medulla. Quantitatively, the most important loss of urea from the inner medullary interstitium is thought to occur via the vasa recta,⁴¹⁹ but urea recycling pathways play a major role in limiting the loss of urea from the inner medulla. Three major urea recycling pathways are described later in this section, and an overview of these is shown in Fig. 10.20).

1. Recycling of Urea Through the Ascending Limbs, Distal Tubules, and Collecting Ducts

Urea that escapes the inner medulla in the ascending limbs of the long loops of Henle is carried back through the thick ascending limbs, distal convoluted tubules, and early portions of the collecting duct system by the flow of tubule fluid.⁴²⁰ When it reaches the urea-permeable part of the inner medullary collecting ducts, it passively exits into the inner medullary interstitium and starts the cycle again.

2. Recycling of Urea Through the Vasa Recta, Short Loops of Henle, and Collecting Ducts

The delivery of urea to the superficial distal tubule exceeds the delivery out of the superficial proximal tubule.^{420–422} This

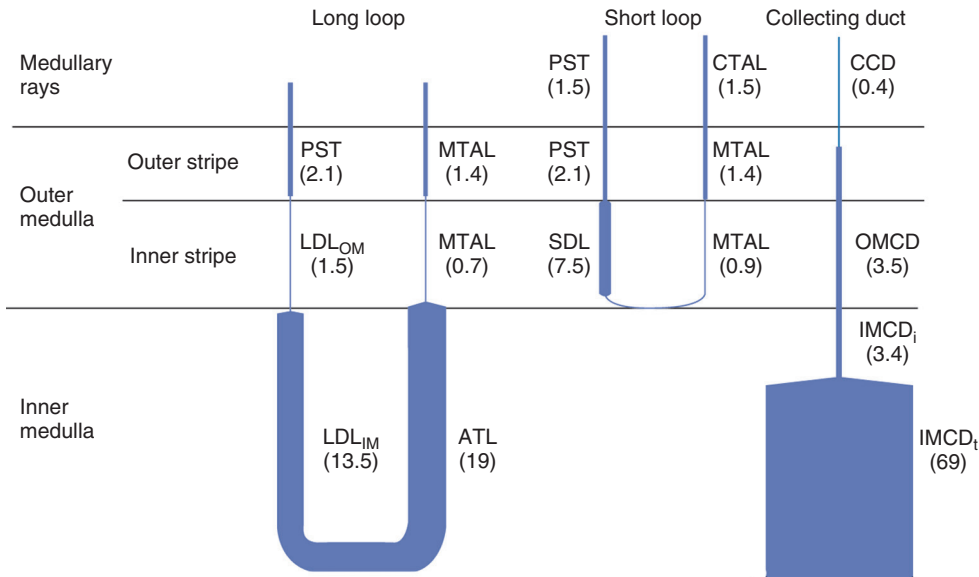


Fig. 10.17 Urea permeabilities of mammalian renal tubule segments. The width of each segment in the diagram is distorted to be proportional to the urea permeability of that segment. Numbers in parentheses are measured values for the permeability coefficient ($\times 10^{-5}$ cm/sec). Values are from isolated perfused tubule studies. *ATL*, Ascending thin limb; *CTAL*, cortical thick ascending limb; *IMCD_i*, initial inner medullary collecting duct; *IMCD_t*, terminal inner medullary collecting duct; *LDL*, thin descending limb of long-looped nephron; *MTAL*, medullary thick ascending limb; *OMCD*, outer medullary collecting duct; *PST*, proximal straight tubule; *SDL*, thin descending limb of short-looped nephron.

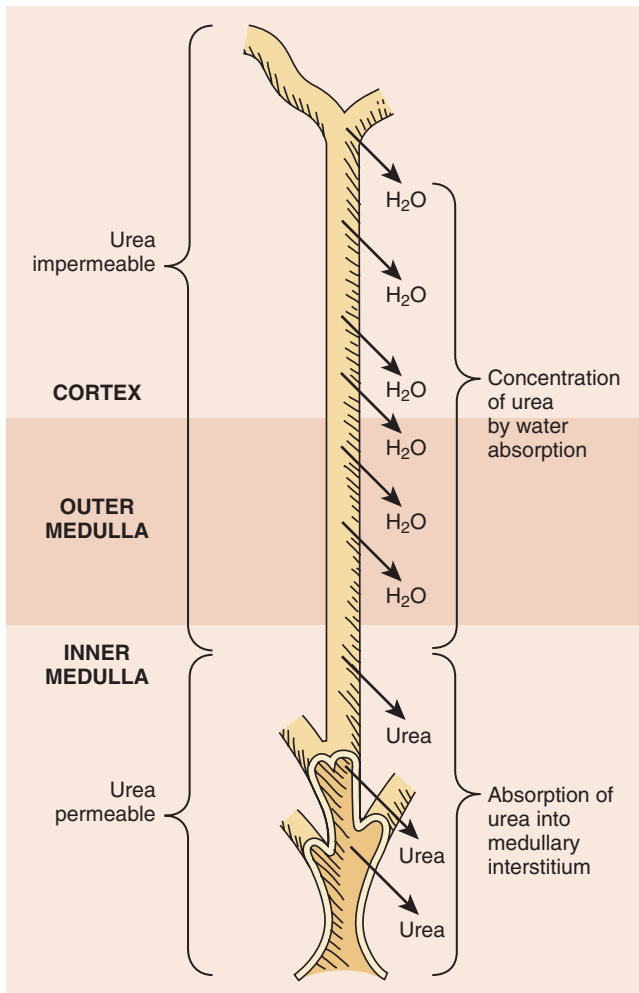


Fig. 10.18 Schematic representation of the mammalian collecting duct system showing principal sites of water absorption and urea absorption. Water is absorbed in the early part of the collecting duct system, driven by an osmotic gradient. Because urea permeabilities of cortical collecting duct, outer medullary collecting duct, and initial inner medullary collecting duct are very low, the water absorption concentrates urea in the lumen of these segments. When the tubule fluid reaches the terminal inner medullary collecting duct, which is highly permeable to urea, urea rapidly exits from the lumen. This urea is trapped in the inner medulla as a result of countercurrent exchange.

implies that net urea addition occurs somewhere along the short loops of Henle. One possible mechanism is that the urea leaving the inner medulla in the vasa recta is transferred to the descending limbs of the short loops of Henle⁴²¹ and is subsequently carried through the superficial distal tubules back to the urea-permeable part of the inner medullary collecting ducts, where it passively exits, completing the recycling pathway. The close physical association between the vasa recta and the descending limbs of the short loops in the vascular bundles of the inner stripe of the outer medulla would facilitate this transfer of urea from the vasa recta to the short loops of Henle.^{26,423} Furthermore, the existence of a facilitative UT, UT-A2, in the thin descending limb of short loops of Henle^{8,371} provides further support for this mechanism. However, as discussed earlier, recent studies on UT-A2

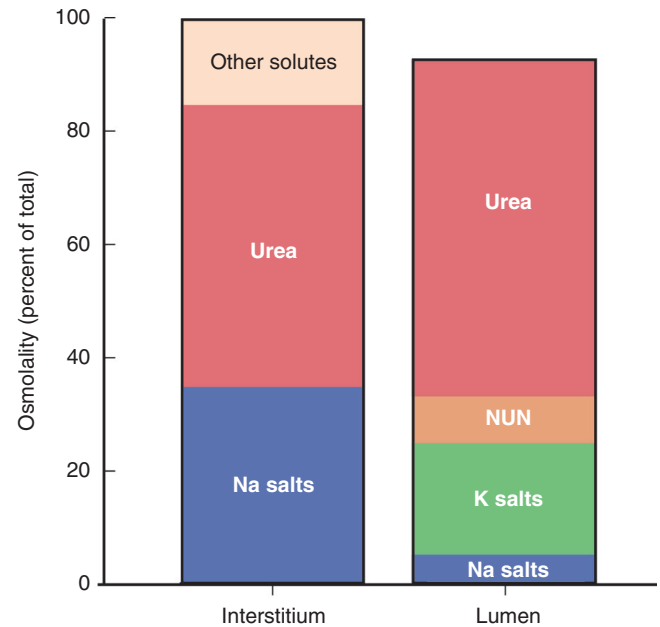


Fig. 10.19 Solutes that account for osmolality of medullary interstitium and tubule fluid in the inner medullary collecting duct during antidiuresis in rats. Urea nearly equilibrates across the inner medullary collecting duct epithelium as a result of rapid facilitated urea transport. Although the osmolalities of the fluid in the two spaces are nearly equal, the nonurea solutes can differ considerably between the two compartments. Typical values in untreated rats are presented. Values can differ considerably in other species and in the same species with different diets. NUN, Nonurea nitrogen.

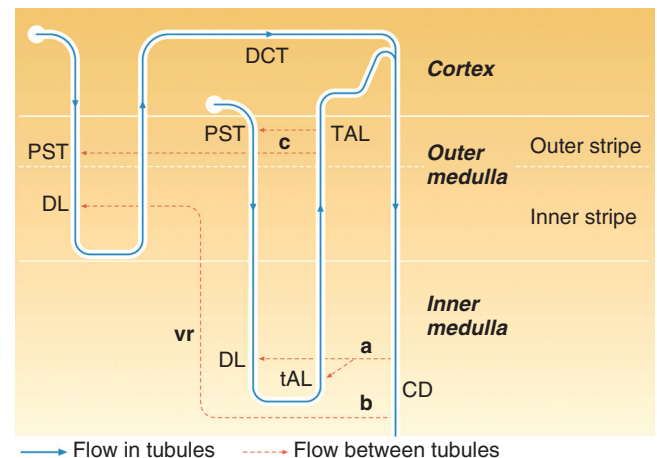


Fig. 10.20 Pathways of urea recycling in renal medulla. Solid blue lines represent a short-looped nephron (left) and a long-looped nephron (right). Transfer of urea between nephron segments is indicated by dashed red arrows labeled a, b, and c corresponding to recycling pathways described in the text. CD, Collecting duct; DCT, distal convoluted tubule; DL, descending limb; PST, proximal straight tubule; tAL, thin ascending limb; TAL, thick ascending limb; vr, vasa recta. Reproduced with permission from Knepper MA, Roch-Ramel F. Pathways of urea transport in the mammalian kidney. *Kidney Int.* 1987;31: 629–633.

knockout mice and UT-A2/UT-B knockout mice have raised doubts about the importance of this pathway.^{5,359}

3. Urea Recycling Between Ascending and Descending Limbs of the Loops of Henle

The urea permeability of thick ascending limbs from the inner stripe of the outer medulla is low.^{424,425} However, the urea permeability of thick ascending limbs from the outer stripe of the outer medulla and the medullary rays is relatively high.^{424,426} Based on this, a urea recycling pathway has been proposed in which urea is reabsorbed from thick ascending limbs and is secreted into neighboring proximal straight tubules, forming a recycling pathway between the ascending limb and descending limbs of the loop of Henle.^{3,419} Urea recycling from the thick ascending limbs and the proximal straight tubules is facilitated by the parallel relationship of these two structures in the outer stripe of the outer medulla and in the medullary rays. This transfer of urea is also likely to depend on a relatively attenuated effective blood flow in these regions. Urea secretion into the proximal straight tubules can occur by passive diffusion,⁴²⁶ active transport,⁴²⁷ or a combination of both. Urea presumably enters the proximal straight tubules of both short- and long-looped nephrons. The urea that enters the short-looped nephrons will be carried back to the inner medulla by the flow of tubule fluid through the superficial distal tubules and cortical collecting ducts, reentering the inner medullary interstitium by reabsorption from the terminal inner medullary collecting duct. The urea that enters proximal straight tubules of long-looped nephrons returns to the inner medulla directly through the descending limbs of the loops of Henle.⁴¹⁹

URINE CONCENTRATION AND DILUTION PROCESSES ALONG THE MAMMALIAN NEPHRON

SITES OF URINE CONCENTRATION AND DILUTION

Micropuncture studies of the mammalian nephron have determined the major sites of tubule fluid concentration and dilution (see Fig. 10.10). Regardless of whether the kidney is diluting or concentrating the urine, proximal tubule fluid is always isosmotic with plasma.⁴²⁸ Whereas early distal convoluted tubule fluid is always hypotonic, the earliest nephron segment where significant differences in tubule fluid osmolality can be detected is the late distal tubule. During water diuresis, the fluid in the distal tubule remains hypotonic. During antidiuresis, the fluid in the distal tubule becomes isosmotic with plasma, and the osmolality between the end of the late distal tubule and the inner medullary collecting ducts rises to a level greater than that of plasma. Thus, the conclusion from micropuncture studies is that the loop of Henle is the major site of dilution of tubule fluid, and that dilution processes in the loop occur regardless of whether the final urine is dilute or concentrated. Further dilution of the tubule fluid can occur in the collecting ducts during water diuresis.⁴²⁹ In contrast, the chief site of urine concentration is beyond the distal tubule (i.e., in the collecting duct system). The mechanisms of urinary dilution and of urinary concentration are discussed in the subsequent sections.

MECHANISMS OF TUBULE FLUID DILUTION

Micropuncture studies in rats show that the fluid in the early distal tubule is hypotonic, due mainly to a reduction in luminal NaCl concentration relative to that in the proximal tubule.⁴³⁰ The low luminal NaCl concentration could result either from active NaCl reabsorption from the loop of Henle or from water secretion into the loop of Henle. Micropuncture measurements in rats, performed using inulin as a volume marker, demonstrate net water reabsorption from the superficial loops of Henle during antidiuresis, thereby ruling out water secretion as a potential mechanism of tubule fluid dilution.⁴²⁰ Thus, one can conclude that luminal dilution occurs because of NaCl reabsorption from the loops of Henle, in excess of water reabsorption. Classic studies of isolated perfused rabbit thick ascending limbs established the mechanism of tubule fluid dilution.^{431,432} NaCl is rapidly reabsorbed by active transport, which lowers the luminal osmolality and NaCl concentration to levels below those in the peritubular fluid. The osmotic water permeability of the thick ascending limb is very low, which prevents dissipation of the trans-epithelial osmolality gradient by water flux.

The tubule fluid remains hypotonic throughout the distal tubule and collecting duct system during water diuresis, aided by the low osmotic water permeability of the collecting ducts when circulating levels of vasopressin are low. Even though the tubule fluid remains hypotonic in the collecting duct system, the solute composition of the tubule fluid is modified within the collecting duct, mainly by Na⁺ absorption and K⁺ secretion. Active NaCl reabsorption from the collecting duct results in a further dilution of the collecting duct fluid, beyond that achieved in the thick ascending limbs.⁴²⁹

MECHANISM OF TUBULE FLUID CONCENTRATION

When circulating vasopressin levels are high, net water absorption occurs between the late distal tubule and the collecting ducts.⁴²⁰ Because water is absorbed in excess of solutes, with a resulting rise in osmolality along the collecting ducts toward the papillary tip,⁴³³ it can be concluded that collecting duct fluid is concentrated chiefly by water absorption, rather than by solute addition.

An axial osmolality gradient in the renal medullary tissue, with the highest degree of hypertonicity at the papillary tip, provides the osmotic driving force for water absorption along the collecting ducts. This osmolality gradient was initially reported by Wirz and colleagues.⁴³⁴ In a classic study, they demonstrated, in antidiuretic rats, the existence of a continuously increasing osmolality gradient along the outer and inner medulla, with the highest osmolality in the deepest part of the inner medulla, the papillary tip. In addition, within the medulla the osmolality of the collecting ducts was as high as in the loops of Henle and the osmolality of vasa recta blood, sampled from near the papillary tip, was virtually equal to that of the final urine.⁴³⁴ Taken together these results demonstrate that the high tissue osmolality was not simply a manifestation of a high osmolality in a single structure, namely, the collecting duct. Micropuncture studies by Gottschalk and Mylle⁴²⁸ based on the superficial and thus accessible tubules and vessels confirmed that the osmolality of the fluid in the loops of Henle, the vasa recta, and the collecting ducts is approximately the same (see Fig. 10.10); thus, these studies

support the hypothesis that the collecting duct fluid is concentrated by osmotic equilibration with a hypertonic medullary interstitium. Furthermore, *in vitro* studies demonstrated that collecting ducts have a high water permeability in the presence of vasopressin,^{99,416} as is required for osmotic equilibration. The mechanism by which the corticomedullary osmolality gradient is generated is considered later.

Although the final axial osmolality gradient within the renal medulla is due to the combined gradients of several individual solutes, as initially demonstrated using tissue slice analysis by Ullrich and Jarausch,⁴³⁵ the principal solutes responsible for the osmolality gradient are NaCl and urea (Fig. 10.21). The increase in the NaCl concentration gradient along the corticomedullary axis occurs predominantly in the outer medulla, with only a small increase in the inner medulla. In contrast, the increase in urea concentration occurs predominantly in the inner medulla, with little or no increase in the initial outer medulla. The mechanisms for generating the NaCl gradient in the outer medulla and urea accumulation in the inner medulla are discussed later.

GENERATION OF THE AXIAL NaCl GRADIENT IN THE RENAL OUTER MEDULLA

In both diuresis and antidiuresis, an osmolality gradient is maintained along the corticomedullary axis of the outer medulla (see Fig. 10.21).⁴³⁶ That gradient arises mostly from an accumulation of NaCl and is generated by the concentrating mechanism of the outer medulla. Because the axial osmolality gradient is present in both diuresis and antidiuresis (in which the outer medullary collecting duct is water-permeable to varying degrees), the accumulation of NaCl in the outer medulla cannot depend on a sustained osmolality difference across the collecting duct epithelium. Thus, the concentrating mechanism must depend on the loops of Henle, on the vasculature, and on their interactions within the outer medulla. Moreover, a mass balance of water and NaCl must be maintained. Thus, for example, concentrated fluid that flows into the inner medulla must be balanced by dilute fluid that, in the presence of vasopressin, is absorbed from the cortical collecting duct, dilutes the cortical interstitial fluid, enters the cortical vasculature, and thus participates in maintaining an appropriate systemic level of blood plasma osmolality.

It has long been believed that the osmolality gradient of the outer medulla is generated by means of countercurrent multiplication of a single effect (“Vervielfachung des Einzeleffektes”). In this paradigm, proposed by Kuhn and Ryffle in 1942,⁴³⁷ osmotic pressure is raised along parallel but opposing flows in nearby tubes that are made contiguous by a hairpin turn (Fig. 10.22); a transfer of solute from one tubule to another (i.e., a single effect) would augment (multiply), or reinforce, the osmotic pressure in the parallel flows. Thus, by means of the countercurrent configuration, a small transverse osmotic difference would be multiplied into a relatively large difference along the axes of flow. In support of this paradigm, Kuhn and Ryffle provided both a mathematical model and an apparatus that exemplified countercurrent multiplication.

As anatomic and physiologic understanding of the renal medulla increased, the countercurrent multiplication paradigm was reinterpreted and modified. In 1951, Hargitay and Kuhn⁴³⁸ put the paradigm in the context of specific renal tubules. The loop of Henle was identified with the parallel

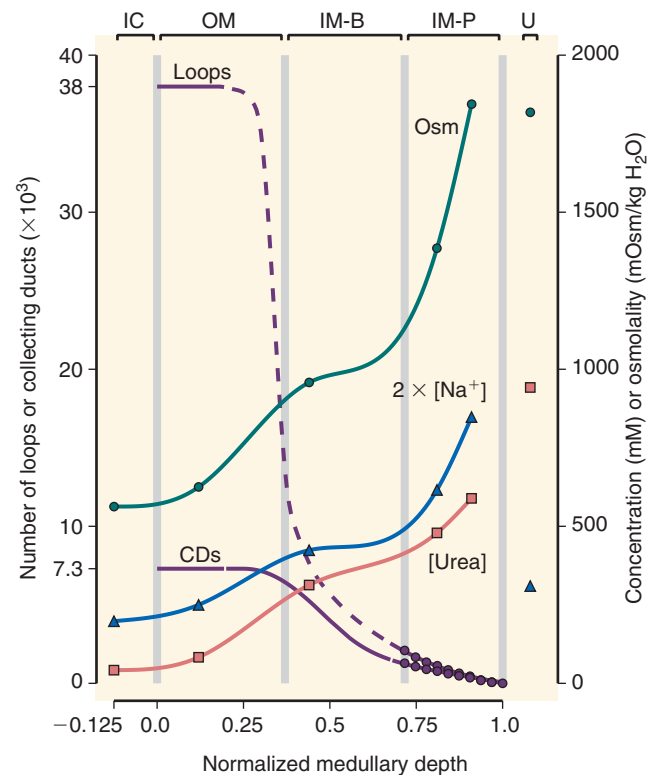


Fig. 10.21 Data from rat kidney in an antidiuretic state. Osmolality, urea concentration, and sodium concentration plus its anion are shown (scale at right), in addition to the loop of Henle and collecting duct populations (scale at left). Loop of Henle and collecting duct populations decrease in inner medulla because collecting ducts merge and loops turn back. The osmolality gradient is larger in the outer medulla and papilla than in the outer part of the inner medulla. The gradient is largest in the papilla, where the osmolality and concentration profiles appear to increase exponentially. The shape of the sodium profile has been corroborated by electron microprobe measurements.⁴⁸⁶ IC, inner cortex; IM, outer part (base) of inner medulla; OM, outer medulla; P, papilla or inner part (tip) of inner medulla; U, urine. Figure based on published data. Curves connecting data points are natural cubic splines, computed by standard algorithms.⁴⁸⁷ Dashed curve segments are interpolations without supporting measurements. Tubule populations in papilla are from Hans et al.⁴⁸⁸ tubule populations in outer medulla are based on estimates in Knepper et al.⁵⁹ Concentrations and osmolalities are from tissue slices and urine samples collected 4.5 hours after onset of vasopressin infusion at 15 μ U/min per 100 g body weight. Data are from Fig. 5 in Atherton et al.⁴⁸⁹ and Figs. 1, 3, 9 in Hai and Thomas⁴³⁶; slice locations were given in Wade et al.⁸ The osmolality reported in the inner cortex seems high relative to the reported plasma concentration of 314 mOsm/kg H₂O. The osmolality and concentration profiles, as drawn in the study by Hai and Thomas,⁴³⁶ apparently do not take into account relative distances between tissue sample sites. (From Sands JM, Layton HE. The urine-concentrating mechanism and urea transporters. In: Alpern RJ, Caplan MJ, Moe OW, eds. *The Kidney: Physiology and Pathophysiology*, ed 5. San Diego: Academic Press; 2013:1463–1510.)

tubes joined by a hairpin turn as proposed by Kuhn and Ryffle. Thus, the loops of Henle were proposed as the source of the outer medullary gradient, and that gradient was hypothesized to draw water out of water-permeable collecting ducts. In 1959, Kuhn and Ramel⁴³⁹ used a mathematical model to show that active transport of NaCl from thick ascending

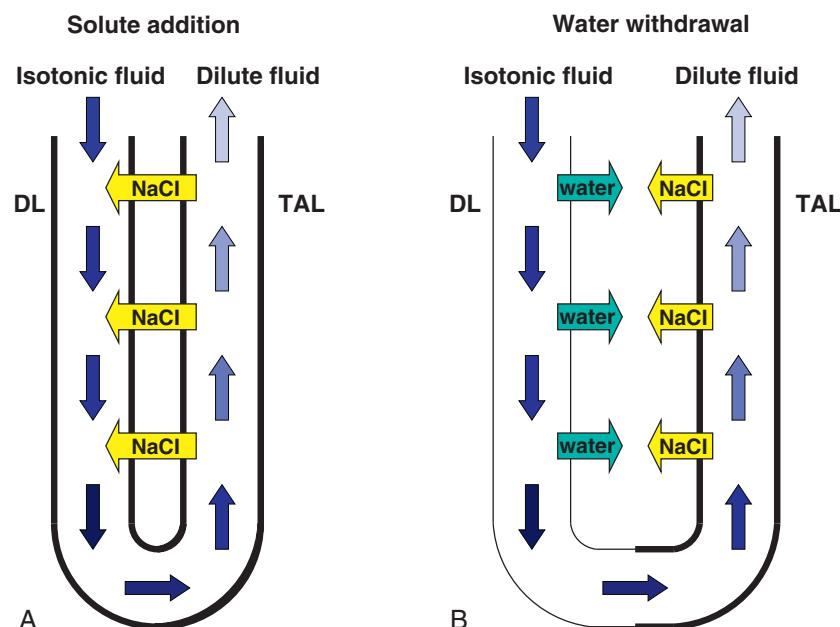


Fig. 10.22 (A) Countercurrent multiplication by means of NaCl transfer from an ascending flow to a descending flow. (B) Countercurrent multiplication by means of water withdrawal from a descending flow. NaCl transport from the ascending flow into the interstitium raises interstitial osmolality; this results in passive water transport from the descending flow, which has lower osmolality than the interstitium. In both panels, tubular fluid flow direction is indicated by blue arrows; increasing osmolality is indicated by darkening shades of blue. Ascending flow may be considered to be in the thick ascending limb of Henle (TAL); descending flow in the descending limb of Henle (DL). Thick black lines indicate that a tubule is impermeable to water; thin lines indicate high permeability to water. (From Layton AT, Layton HE. Countercurrent multiplication may not explain the axial osmolality gradient in the outer medulla of the rat kidney. *Am J Physiol Renal Physiol.* 2011; 301:F1047-F1056.)

limbs could serve as the single effect. Subsequent physiologic experiments confirmed the active NaCl transport and the osmotic absorption of water from collecting ducts.^{420,431–433} Experiments indicating high water permeability in hamster descending limbs of short loops⁴⁴⁰ and in descending limbs of long loops^{9,10,12} suggested that the accumulation of NaCl from thick limbs concentrated descending limb tubular fluid by osmotic water withdrawal, rather than by NaCl addition (see Fig. 10.22).

In more recent years, as anatomic details of the medulla emerged, it has become necessary to refine the paradigm of countercurrent multiplication to provide an accurate representation of the means by which the gradient is generated in the mammalian outer medulla. In particular, the descending limbs of short loops have been shown to be anatomically separated from ascending limbs, with inner stripe portions of short loops near (or within) the vascular bundles and thick limbs near the collecting ducts.^{23,441} This configuration is not consistent with direct interactions between counterflowing limbs. Furthermore, in short-looped rat nephrons, Wade et al.⁸ found that AQP1 is not expressed in portions of descending limb segments in the distal inner stripe. Zhai et al.¹⁷ found that AQP1 is not expressed in descending limbs of short loops in the inner stripes of mice, rats, and humans. The absence of AQP1 suggests that the assumption of high water permeability in descending limbs of short loops merits further experimental study.

From these considerations, it seems reasonable to hypothesize that the outer medullary osmolality gradient arises principally from vigorous active transport of NaCl,

without accompanying water, from the thick ascending limbs of short- and long-looped nephrons. The tubular fluid of the thick limbs that enters the cortex is diluted well below plasma osmolality, and thus the requirement of mass balance is met. In rats and mice, the thick limbs are localized near the collecting ducts⁴⁴²; mathematical models suggest that at a given level of the outer medulla, the interstitial osmolality will be higher near the collecting ducts than near the vascular bundles.^{443,444} This higher osmolality will facilitate water withdrawal from the descending limbs of long loops and from collecting ducts. Descending vasa recta are thought to be found only in the vascular bundles. Thus, the ascending vasa recta will act as the collectors of any NaCl that is absorbed from the loops of Henle and water that is absorbed from the descending limbs of long loops and from collecting ducts.

The countercurrent configuration of the ascending vasa recta, relative to the descending limbs and collecting ducts, is likely to participate in sustaining the axial gradient: as ascending vasa recta fluid ascends toward the cortex, its osmolality will exceed that in the descending limbs of long loops and in the collecting ducts. Thus, ascending vasa recta fluid will be progressively diluted as that fluid contributes to the concentrating of fluid in descending limbs of long loops and in collecting ducts, by giving up NaCl to, and absorbing water from, the interstitium (Fig. 10.23).

The previous summary appears to account for the elevation of osmolality in the outer medulla without invoking a role for countercurrent multiplication. However, a question remains: why does the osmolality gradient increase along the outer

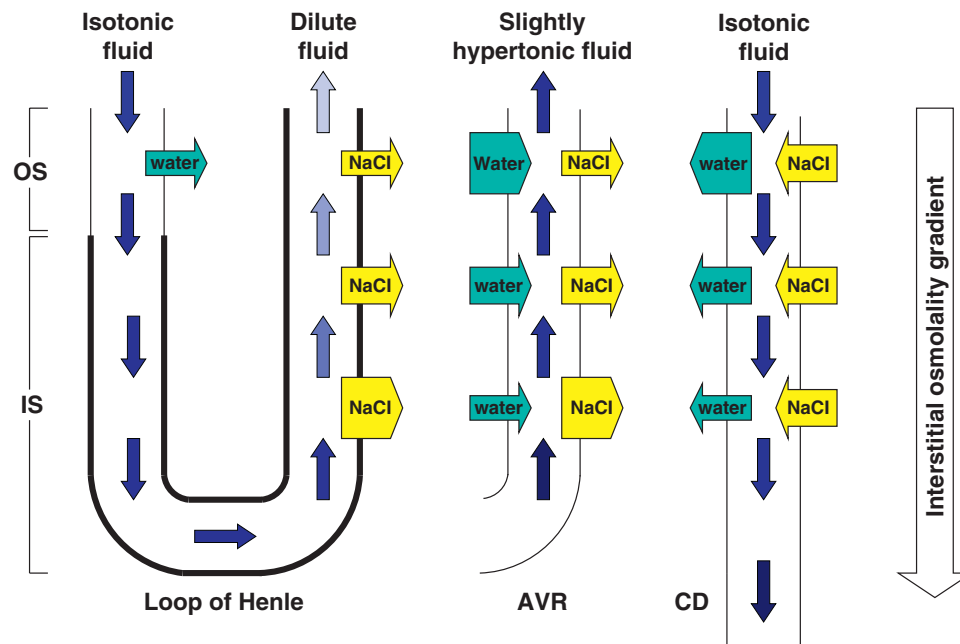


Fig. 10.23 Outer medullary concentrating mechanism based on NaCl addition to the interstitium but without water absorption from descending limbs of short loops. Arrows indicate water (cyan) and NaCl (yellow) transepithelial transport; arrow widths suggest relative transport magnitudes. Isotonic fluid is considered to have the same osmolality as blood plasma. Flow entering the AVR is assumed to arise from a descending vas rectum that is in, or near, a vascular bundle. Outflow from the collecting duct enters the inner medullary collecting duct. Tubular fluid flow direction is indicated by blue arrows; increasing osmolality is indicated by darkening shades of blue. Thick black lines indicate that a tubule is impermeable to water; thin lines indicate high permeability to water. AVR, Ascending vasa recta; CD, collecting duct; IS, inner stripe; OS, outer stripe. (From Layton AT, Layton HE. Countercurrent multiplication may not explain the axial osmolality gradient in the outer medulla of the rat kidney. *Am J Physiol Renal Physiol.* 2011; 301:F1047-F1056.)

medulla as a function of increasing medullary depth? The answer likely lies in the local balance of NaCl absorption from thick limbs and water absorption from descending limbs of long loops and from collecting ducts. At deeper medullary levels, the rate of NaCl absorption from thick limbs may be higher than at shallow levels, owing to a higher Na-K-ATPase activity at deeper levels⁴⁴⁵ and to a saturation of transport proteins by the higher NaCl concentration in thick limb tubular fluid before dilution. Moreover, because of the water already absorbed in the upper outer medulla, the load of water presented to the thick limbs deep in the outer medulla by descending limbs of long loops and by the collecting ducts is much reduced.

A caveat is in order: our understanding of the outer medulla is mostly based on information obtained from heavily studied laboratory animals, especially rats and mice. Outer medullary function and structure are likely to vary substantially in other species. For example, the human kidney has limited concentrating capability (relative to many other mammals) and only about one seventh of the loops of Henle are long⁴⁴⁶; the mountain beaver (*Aplodontia rufa*) has mostly cortical loops of Henle and essentially no inner medulla.⁴⁴⁷ It seems likely that the outer medullary structure in these species differ substantially from that in rats and mice. Finally, it should be acknowledged that the paradigm formulated above is similar to that proposed by Berliner et al. in 1958.⁸³

DETERMINANTS OF CONCENTRATING ABILITY

The overall concentrating ability of the kidney arises from interactions among several differing components. In addition

to the active transport of NaCl from the thick ascending limbs and the water permeability of the collecting ducts, two other factors play a significant role in determining the osmolality of the final urine. One important determinant is the delivery rate of NaCl and water to the loop of Henle, which sets an upper limit on the amount of NaCl actively reabsorbed by the thick ascending limb to drive the outer medullary concentrating mechanism. Another important determinant is the volume of tubular fluid delivered to the medullary collecting duct, which has an underappreciated effect on the concentrating process. Too much fluid delivery saturates water reabsorption processes along the medullary collecting ducts, leading to interstitial dilution due to rapid osmotic water transport. In contrast, too little fluid delivery to the medullary collecting ducts, even in the absence of vasopressin, results in sustained osmotic equilibration across the collecting duct epithelium owing to the nonzero osmotic water permeability of the inner medullary collecting duct.^{144,148,149}

AN UNRESOLVED QUESTION: CONCENTRATION OF NaCl IN THE RENAL INNER MEDULLA

Tissue slice studies demonstrate that the corticomedullary osmolality gradient is made up largely of a NaCl gradient in the outer medulla and a urea gradient in the inner medulla (see Fig. 10.21). Accordingly, in the previous sections we have emphasized the processes that concentrate NaCl in the outer medulla and the processes responsible for urea accumulation in the inner medulla (passive urea absorption from the inner medullary collecting duct plus countercurrent exchange of urea via diffusion). The concentrating

mechanism described previously functions only in the renal outer medulla and medullary rays of the cortex. The ascending limbs of loops of Henle that reach into the inner medulla are thin-walled and do not actively transport NaCl^{416,448,449}; nonetheless, in antidiuresis a substantial axial osmolality gradient is generated in the inner medulla of many mammals. For nearly 50 years, controversy has persisted regarding the nature of the mechanism that generates the inner medullary osmolality gradient. Moreover, the energy source for the concentrating of nonurea solutes in the inner medullary interstitium is not known. General analysis of inner medullary concentrating processes indicates that, to satisfy mass balance requirements, either an ascending stream (thin ascending limbs or ascending vasa recta) must be diluted relative to the inner medullary interstitium, or a descending stream (descending thin limbs, descending vasa recta, or collecting ducts) must be concentrated locally relative to the inner medulla.^{87,450}

Three major hypotheses have been proposed for the concentrating mechanism of the inner medulla.

1. The “Passive Mechanism”

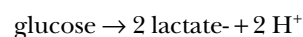
Kokko and Rector⁴⁰⁸ and Stephenson⁴⁰⁹ independently proposed a model by which the osmolality in the thin ascending limb could be lowered below that of the interstitium entirely by passive transport processes in the inner medulla. This mechanism is generally referred to as the “passive model” or the “passive countercurrent multiplier mechanism.” The passive mechanism depends on the separation of urea and NaCl that is accomplished by NaCl absorption from the thick ascending limbs; indeed, this absorption is the hypothesized energy source for the passive mechanism. In this model, rapid urea reabsorption from the inner medullary collecting duct generates and maintains a high urea concentration in the inner medullary interstitium, causing the osmotic withdrawal of water from the thin descending limb. This concentrates NaCl in the descending limb lumen and results in a transepithelial gradient favoring the passive reabsorption of NaCl from the thin ascending limb of Henle loop. Additionally, if the ascending limbs have extremely low urea permeability, then any NaCl that has been reabsorbed from the thin ascending limb will not be replaced by urea. Thus, the ascending limb fluid will be dilute relative to the fluid in other nephron segments, generating a “single effect” analogous to active NaCl absorption from thick ascending limbs. This single effect can then be multiplied by the counterflow between the ascending and descending limbs of Henle loops. This model requires that the thin descending limbs are highly permeable to water but not NaCl or urea, whereas the thin ascending limb would have to be permeable to NaCl but not water or urea. However, several objections to the passive mechanism have since been made. Contrary to the permeability requirements of the passive model, high urea permeabilities have been measured in the thin descending limb and thin ascending limb (summarized in Gamba and Knepper⁴⁵¹), whereas little or no osmotic water permeability has been measured in the lower portions of thin descending limbs in the inner medulla.²⁰ In addition, studies in UT-A1/A3 urea transporter knockout mice found that urea accumulation in the inner medulla was largely eliminated, but inner medullary NaCl accumulation was not affected^{143,358,407} (see “UT-A1/A3 knockout mice” in earlier section).

Layton and colleagues reevaluated the passive mechanism by incorporating measured loop NaCl, urea, and water permeabilities into a mathematical model.^{19,452,453} These studies suggest that water absorption from descending limbs is not a requirement for the passive mechanism to generate an osmolality gradient, and that the urea-permeable loops of Henle can serve as a highly effective countercurrent urea exchanger. However, the model was able to fully account for the high urine osmolalities attained by some animals.

2. Concentrating Mechanism Driven by External Solute

Jen and Stevenson⁴⁵⁴ proposed that the concentrating mechanism of the inner medulla depends on a solute other than NaCl and urea. By means of a mathematical model, they demonstrated, in principle, that the continuous addition of small amounts of an unspecified, but osmotically active, solute to the inner medullary interstitium could produce a substantial axial osmolality gradient. Such a solute would have to be generated in the inner medulla by a chemical reaction that produces more osmotically active particles than it consumes. The mechanism of concentration is similar to that driven by urea in the “passive” models proposed by Kokko and Rector and by Stephenson^{408,409}: the thin descending limbs in the inner medulla are assumed impermeable to the solute (thus it is an “external” solute), and as a result, water is withdrawn from the descending limbs and the concentration of NaCl is raised in descending limb tubular fluid. Beginning at the loop bend, elevated NaCl concentration within the loop will result in a substantial NaCl efflux that will dilute the ascending flow and that is sufficient to generate the axial gradient.

The feasibility of this mechanism was subsequently confirmed by Thomas and Wexler⁴⁵⁵ in the context of a more detailed mathematical model. In further modeling studies, Thomas,⁴⁵⁶ and Hervy and Thomas⁴⁵⁷ proposed that lactate, generated by anaerobic glycolysis (the predominant means of ATP generation in the inner medulla), could serve as the solute. Two lactate ions are generated per glucose consumed:



However, as pointed out by Knepper et al.,⁸⁷ the net generation of osmotically active particles depends on which buffering anions are titrated by the protons. If the protons titrate bicarbonate, there may be a net removal of osmotically active particles; if instead the protons titrate other buffers (e.g., phosphate or NH_3), there will be a net generation of osmotically active particles. Mathematical models developed by Zhang and Edwards⁴⁵⁸ and by Chen et al.⁴⁵⁹ predicted that vascular countercurrent exchange would tend to restrict significant glucose availability into the outer medulla and the upper inner medulla, thus limiting the rate of lactate generation in the deep inner medulla where the highest osmolalities are found.

3. Hyaluronan as a Mechano-osmotic Transducer

In the mechano-osmotic induction hypothesis,^{87,460} in which energy from the peristaltic contractions of the renal pelvic wall is used to concentrate solutes in the descending limbs and collecting ducts by water withdrawal, or, alternatively, the peristaltic contractions reduce sodium activity in the hyaluronan matrix of the interstitium, resulting in the

reabsorption of hypotonic fluid from that matrix into ascending vasa recta. Hyaluronan (or “hyaluronic acid”) is a glycosaminoglycan^{461,462} that is abundant in the interstitium of the renal inner medulla.^{463,464} The hyaluronan in the inner medulla is produced by a specialized interstitial cell (the type 1 interstitial cell), which forms characteristic “bridges” between the thin limbs of Henle and the vasa recta.⁴⁶⁵ These bridges may delimit, above and below, the nodal compartments identified by Pannabecker and Dantzer,²⁸ which were discussed previously. Thus, the inner medullary interstitium may be considered to be composed of a compressible, viscoelastic, hyaluronan matrix.

Several hypotheses have been advanced that depend on the peristalsis of the papilla as an integral component of the concentrating mechanism of the inner medulla.^{87,357,466} In one hypothesis, which was suggested in part by Schmidt-Nielsen,⁴⁶⁷ compression of the hyaluronan matrix stores some of the mechanical energy from the smooth muscle contraction that gives rise to the peristaltic wave. In the postwave decompression, the matrix exerts an elastic force that promotes water absorption from thin descending limbs and collecting ducts, and thereby increases tubular fluid osmolality. Water absorption from the descending limbs would raise tubular fluid NaCl concentration and thus promote a vigorous NaCl absorption from the loop bends and early ascending limbs. However, if, as is apparently the case in rat, the lower 60% of inner medullary descending limbs are water impermeable,²⁰ water is unlikely to be absorbed from descending limbs in the deep portion of the inner medulla where the highest osmolalities are achieved.

Another hypothesis involves special properties of hyaluronan.⁴⁶⁸ Hyaluronan is a large polyanion (1000 to 10,000 kDa). Its charge is due to the carboxylate (COO) groups of the glucuronic subunits. Hyaluronan is hydrophilic and assumes a highly expanded, random coil conformation that occupies a large volume of space relative to its mass. This extended state arises partly from electrostatic repulsion between carboxylate groups (which maximize the distances between neighboring negative charges), and partly from the extended conformations of the glycosidic bonds.

Knepper et al. proposed that the periodic compression of the papilla, and the effects of that compression on the hyaluronan matrix, could explain the osmolality gradient along the inner medulla.⁸⁷ When hyaluronan is compressed, the repulsive forces of neighboring carboxylate groups are overcome, in part, by a condensation of cations (mainly Na⁺), and a localized crystalloid structure is formed. Thus, compression of the hyaluronan gel results in a decrease of the local sodium ion activity in the gel.⁸⁷ In aqueous solutions that are in equilibrium with the gel, the NaCl concentration will decrease as a consequence of the compression-induced reduction in Na⁺ activity within the gel. Therefore, the free fluid that is expressed from the hyaluronan matrix during the contraction phase will have a lower total solute concentration than that of the gel as a whole. The slightly hypotonic fluid expressed from the matrix is likely to escape the inner medulla via the ascending vasa recta, the only structure that remains open during the compressive phase of the contraction cycle.⁴⁶⁷ As a consequence, the ascending fluid within the ascending vasa recta would have a lower osmolality than the local interstitium, and as a result, fluid in collecting ducts and descending vasa recta would be concentrated.

This mechanism is consistent with the nodal compartments found by Pannabecker and Dantzer²⁸; these compartments, which are likely rich in hyaluronan, are in contact with collecting ducts, thin ascending limbs, and ascending vasa recta. Thus, they are well-configured to be sites of transduction (i.e., sites where the mechanical energy of peristalsis is harnessed to generate an ascending flow that is dilute relative to average local osmolality). However, no quantitative analyses or mathematical models have examined the mass balance consistency or the thermodynamic adequacy of hypotheses that depend on the peristaltic contractions.

ACKNOWLEDGMENTS

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Drs. Dennis Brown and Robert A. Fenton were coauthors of a chapter on the cell biology of vasopressin action in the 10th edition and some of the material in that chapter is incorporated into this chapter in the present edition.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Which diuretic will interfere with the ability to both concentrate and dilute the urine?
- Thiazide
 - Furosemide
 - Amiloride
 - Acetazolamide
 - Spironolactone

Answer: b

Rationale: Furosemide inhibits NKCC2 in the thick ascending limb, thereby inhibiting NaCl reabsorption. The inability to remove NaCl from the luminal fluid interferes with the ability to generate dilute urine. The inability to add NaCl to the medullary interstitium prevents the generation of a hypertonic medulla and interferes with the ability to concentrate the urine. The other diuretic choices do not act on the thick ascending limb.

2. All of the following are required to reabsorb water across the collecting duct EXCEPT
- Hypertonic medulla
 - Vasopressin (ADH)
 - Aquaporin 1 water channels
 - Aquaporin 2 water channels
 - Aquaporin 3 water channels

Answer: c

Rationale: Vasopressin acts on the collecting duct to stimulate transcellular water reabsorption via aquaporin 2 in the apical membrane and aquaporin 3 in the basolateral membrane, provided that a hypertonic medulla exists to provide the osmotic driving force for water reabsorption. Aquaporin 1 is not expressed in the collecting duct.

3. Which urea recycling pathway is most likely to play a major role in urine concentration?
- Recycling of urea through the ascending limbs, distal tubules, and collecting ducts
 - Recycling of urea through the vasa recta, short loops of Henle, and collecting ducts
 - Urea recycling between ascending and descending limbs of the loops of Henle
 - Urea recycling across the renal pelvic (calyceal) epithelium
 - Urea recycling from the proximal tubule to peritubular capillaries

Answer: a

Rationale: Vasopressin stimulates urea reabsorption across the inner medullary collecting duct. The reabsorbed urea can be recycled into the thin ascending limb in the inner medulla. This recycled urea remains in the tubular fluid as it moves into the urea impermeable thick ascending limb, distal tubule, and cortical collecting duct. The other choices represent urea recycling pathways but are less important than the inner medullary urea recycling pathway in choice a.

4. Urine concentrating ability is improved by
- Hypokalemia
 - Protein malnutrition
 - Hypercalcemia
 - Decrease in medullary blood flow
 - Mutations in the V_2 receptor

Answer: d

Rationale: A decrease in medullary blood flow improves urine-concentrating ability but improving the efficiency of countercurrent exchange while an increase reduces urine concentrating ability by decreasing the efficiency of countercurrent exchange. The other choices all cause reduced urine-concentrating ability.